

Read this book from beginning to the end very carefully, then also read the correction document of this book fully and carefully too. I want to know, whether the equations of motion of chapter 9 are correctly derived in the book or not, so I want to you to carefully check all the derivations related to the EOMs of chapter 9 and derive step-by-step by yourself to double check them.

已思考 3m 57s >

I checked the Chapter 9 derivation chain against the errata. My conclusion is:

The printed book's Chapter 9 EOM derivation is not fully correct as printed. The errata identifies real mistakes in the angular-momentum derivation, Eq. (9.31) and Eq. (9.32), and a vertical-force typo in Eq. (9.44). After applying those errata, the main simplified EOMs, especially Eq. (9.39), Eq. (9.40), and the load-transfer results Eq. (9.45)–(9.49), are consistent with the chapter's assumptions: small roll/pitch, rigid tires, flat road, negligible suspension jacking, and neglect of higher-order products.

SVDerrata

SVDerrata

1. The corrected Chapter 9 EOM core

Let

$$L = h - d$$

where h is CG height and d is the VIP/no-roll-center height used in Chapter 9. The vehicle invariant point M is used to define the vehicle velocity,

$$\mathbf{V}_M = u\mathbf{i}_1 + v\mathbf{j}_1,$$

so

$$\mathbf{a}_M = (\dot{u} - vr)\mathbf{i}_1 + (\dot{v} + ur)\mathbf{j}_1.$$

The CG position relative to M is

$$\mathbf{MG} = L\mathbf{k}_3 \simeq L(\theta\mathbf{i}_1 - \phi\mathbf{j}_1 + \mathbf{k}_1).$$

Using rigid-body kinematics,

$$\mathbf{a}_G = \mathbf{a}_M + \dot{\boldsymbol{\Omega}} \times \mathbf{MG} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{MG}),$$

and keeping only the terms retained by the book gives

$$\mathbf{a}_G \simeq (\dot{u} - vr + L\ddot{\theta})\mathbf{i}_1 + (\dot{v} + ur - L\ddot{\phi})\mathbf{j}_1.$$

Therefore the force equations in Eq. (9.39) are correct:

$$m(\dot{u} - vr + L\ddot{\theta}) = X,$$

$$m(\dot{v} + ur - L\ddot{\phi}) = Y,$$

$$0 = Z.$$

The angular-momentum part gives, after applying the errata and then simplifying consistently,

$$\dot{\mathbf{K}}_G^r \simeq (J_x\ddot{\phi} - J_{xx}\dot{r})\mathbf{i}_1 + J_y\ddot{\theta}\mathbf{j}_1 + (J_z\dot{r} - J_{xx}\dot{\phi})\mathbf{k}_1.$$

The torque equation about M is

$$\dot{\mathbf{K}}_G^r + \mathbf{MG} \times m\mathbf{a}_G = \mathbf{M}_M.$$

The cross product is

$$\mathbf{MG} \times m\mathbf{a}_G \simeq m \left[(L^2\ddot{\phi} - L(\dot{v} + ur))\mathbf{i}_1 + (L^2\ddot{\theta} + L(\dot{u} - vr))\mathbf{j}_1 + L\dot{u}\phi\mathbf{k}_1 \right].$$

So the final simplified EOMs are

$$m(\dot{u} - vr + L\ddot{\theta}) = X,$$

$$m(\dot{v} + ur - L\ddot{\phi}) = Y,$$

$$0 = Z,$$

$$(J_x + mL^2)\ddot{\phi} - J_{xx}\dot{r} - mL(\dot{v} + ur) = L_M,$$

$$(J_y + mL^2)\ddot{\theta} + mL(\dot{u} - vr) = M_M,$$

$$J_z\dot{r} - J_{xx}\dot{\phi} + mL\dot{u}\phi = N.$$

This matches Eq. (9.39), with $L = h - d$.

978-94-017-8533-4

2. The real mistakes in the printed book

Mistake A: Eq. (9.31)

The printed angular-momentum derivative has wrong signs inside the last parentheses of each row. The errata corrects Eq. (9.31). The corrected first row, for example, contains

$$-J_{xx}(\dot{r} + pq),$$

not the printed form with the wrong sign. The same kind of correction is needed in the other rows.

SVDerrata

This is a genuine derivation error. It comes from the cross product

$$\boldsymbol{\Omega} \times \mathbf{K}_G.$$

If the inertia tensor is written with products of inertia as

$$\mathbf{K}_G = (J_xp - J_{xy}q - J_{xz}r)\mathbf{i}_3 + (-J_{xy}p + J_yq - J_{yz}r)\mathbf{j}_3 + (-J_{xz}p - J_{yz}q + J_zr)\mathbf{k}_3,$$

then

$$\dot{\mathbf{K}}_G = \left(\frac{d\mathbf{K}_G}{dt} \right)_{S_1} + \boldsymbol{\Omega} \times \mathbf{K}_G.$$

Doing that cross product gives the errata version, not the printed version.

Mistake B: Eq. (9.32)

After assuming $J_{xy} \simeq J_{yz} \simeq 0$, the first row of Eq. (9.32) should also contain

$$-J_{xx}(\dot{r} + pq),$$

not the printed sign.

SVDerrata

Important point: this error does not change the final Eq. (9.39) at the approximation order used in the chapter, because the affected pq term is later discarded as a higher-order small product. So Eq. (9.31)/(9.32) are wrong as printed, but the final simplified EOMs survive.

Mistake C: Eq. (9.44)

The vertical resultant in Eq. (9.44) is wrong as printed. It should be

$$Z = Z_1 + Z_1^q + Z_2 + Z_2^q - mg,$$

not with Z_1^q repeated twice.

SVDerrata

This is a real force-balance typo. If front and rear aerodynamic vertical loads differ, the printed equation would give the wrong vertical resultant.

3. Load-transfer equations

With the corrected right-hand side, the lateral load-transfer derivation is consistent. The chapter gives

$$\Delta Z_1 t_1 = Y_1 q_1 + k_{\phi 1} \phi + c_{\phi 1} \dot{\phi},$$

$$\Delta Z_2 t_2 = Y_2 q_2 + k_{\phi 2} \phi + c_{\phi 2} \dot{\phi}.$$

Adding them gives

$$\Delta Z_1 t_1 + \Delta Z_2 t_2 = k_\phi \phi + c_\phi \dot{\phi} + Yd.$$

Combining with the roll torque equation gives

$$Y(h - d) + mg(h - d)\phi = k_\phi \phi + c_\phi \dot{\phi} + J_x \ddot{\phi} - J_{xx} \dot{r}.$$

Then the total lateral load-transfer result is

$$\Delta Z_1 t_1 + \Delta Z_2 t_2 = ma_y h + mg(h - d)\phi - (J_x \ddot{\phi} - J_{xx} \dot{r}).$$

The longitudinal load-transfer equation is also consistent:

$$\Delta Z = \frac{-Xh + J_y \ddot{\theta}}{l}.$$

These match the logic of Eq. (9.45)–(9.49).

978-94-017-8533-4

4. Final verdict

As printed: Chapter 9 is not fully correct, because Eq. (9.31), Eq. (9.32), and Eq. (9.44) contain errors.

After applying the errata: the main Chapter 9 EOMs are correct within the chapter's stated modeling assumptions and linearization order.

For your own vehicle model work, I would not copy Chapter 9 as a general 3D multibody EOM. It is a carefully linearized handling-with-roll model, not a full 14DOF road-relative model. But as a reduced roll/pitch/yaw handling model, the corrected equations are mechanically sound.

已思考 3m 53s >

I double-checked it. I would refine my previous answer slightly:

Chapter 9 is not correct as printed, because the errata corrects Eq. (9.31), Eq. (9.32), and Eq. (9.44). But after applying those corrections, the main Chapter 9 roll/pitch/yaw EOM and tyre-load equations are internally consistent under the chapter's assumptions: rigid tyres, flat road, small pitch/roll, negligible suspension jacking, and the Chapter 9 linearization order. The book explicitly says the Chapter 9 roll model reuses the Chapter 3 tyre equations and assumes rigid tyres; the errata corrects the Chapter 9 angular-momentum signs and the repeated Z_1^a in Eq. (9.44).

Below is the corrected full equation set I would use.

1. Definitions

Let

$$L = h - d$$

where h is the CG height and d is the VIP/no-roll-center height used in Chapter 9.

$$r = \psi, \quad l = a_1 + a_2$$

The Chapter 9 longitudinal and lateral accelerations are

$$a_x = (\dot{u} - vr) + L\ddot{\theta}$$

$$a_y = (\dot{v} + ur) - L\ddot{\phi}$$

so that

$$X = ma_x, \quad Y = ma_y.$$

2. Corrected main EOM, negligible unsprung mass

The corrected Chapter 9 EOM are

$$m \left[(\dot{u} - vr) + L\ddot{\theta} \right] = X$$

$$m \left[(\dot{v} + ur) - L\ddot{\phi} \right] = Y$$

$$0 = Z$$

$$(J_x + mL^2) \ddot{\phi} - J_{xz} \dot{r} - mL(\dot{v} + ur) = L_M$$

$$(J_y + mL^2) \ddot{\theta} + mL(\dot{u} - vr) = M_M$$

$$J_z \dot{r} - J_{xz} \dot{\phi} + mL\dot{u}\phi = N_M = N.$$

Equivalently, using $X = ma_x$ and $Y = ma_y$, the last three can be written in the book's compact form as

$$J_x \ddot{\phi} - J_{xz} \dot{r} - ma_y L = L_M$$

$$J_y \ddot{\theta} + ma_x L = M_M$$

$$J_z \dot{r} - J_{xz} \dot{\phi} + ma_x L\phi = N.$$

The first form is the safer one to implement if you want to follow Eq. (9.39) literally; the second is the book's Eq. (9.40)-style rewriting. The difference between $mL\dot{u}\phi$ and $ma_x L\phi$ is within the small-term approximations used in the chapter. The book gives these EOM in Eq. (9.39) and rewrites them in Eq. (9.40).

3. Corrected force and moment right-hand sides

The corrected resultant forces and moments are

$$X = X_1 + X_2 - X_a$$

with, in the book's simplified straight-ahead aerodynamic form,

$$X_a = \frac{1}{2} \rho S C_x u^2.$$

$$Y = Y_1 + Y_2$$

$$Z = Z_1 + Z_1^a + Z_2 + Z_2^a - mg$$

This is the important errata correction: the printed book repeats Z_1^a , but it must be Z_1^a and Z_2^a .

The roll, pitch, and yaw moments are

$$L_M = -k_\phi \phi - c_\phi \dot{\phi} + mgL\phi$$

$$M_M = -k_\theta \theta - c_\theta \dot{\theta}$$

$$N_M = N = N_Y + N_X$$

with

$$N_Y = Y_1 a_1 - Y_2 a_2$$

$$N_X = \Delta X_1 t_1 + \Delta X_2 t_2.$$

Therefore,

$$N = (Y_1 a_1 - Y_2 a_2) + (\Delta X_1 t_1 + \Delta X_2 t_2).$$

The roll and pitch stiffness/damping definitions are

$$k_\phi = k_{\phi 1} + k_{\phi 2}, \quad c_\phi = c_{\phi 1} + c_{\phi 2}$$

$$k_\theta = k_1 a_1^2 + k_2 a_2^2, \quad c_\theta = c_1 a_1^2 + c_2 a_2^2.$$

These are Chapter 9 Eq. (9.42)–(9.44), with the Z -equation corrected by the errata.

4. Fully substituted corrected EOM

Substituting the corrected right-hand sides into the EOM gives the practical final set:

$$m \left[(\dot{u} - vr) + L\ddot{\theta} \right] = X_1 + X_2 - \frac{1}{2} \rho S C_x u^2$$

$$m \left[(\dot{v} + ur) - L\ddot{\phi} \right] = Y_1 + Y_2$$

$$Z_1 + Z_1^a + Z_2 + Z_2^a - mg = 0$$

$$(J_x + mL^2) \ddot{\phi} - J_{xz} \dot{r} - mL(\dot{v} + ur) = -k_\phi \phi - c_\phi \dot{\phi} + mgL\phi$$

$$(J_y + mL^2) \ddot{\theta} + mL(\dot{u} - vr) = -k_\theta \theta - c_\theta \dot{\theta}$$

$$J_z \dot{r} - J_{xz} \dot{\phi} + mL\dot{u}\phi = (Y_1 a_1 - Y_2 a_2) + (\Delta X_1 t_1 + \Delta X_2 t_2).$$

That is the corrected Chapter 9 EOM set I would trust.

已思考 2m 59s >

I re-checked the derivation from the Chapter 9 kinematics, angular momentum, torque equation about the VIP M , the Chapter 9 errata, and the Chapter 3 tyre-load definitions. My revised conclusion is:

Yes: the equations below are the corrected Chapter 9 equations, consistent with the book's assumptions. But no equation should be used directly from the printed book without applying the errata, because Eq. (9.31), Eq. (9.32), and Eq. (9.44) are wrong as printed. The errata explicitly corrects the angular-momentum signs in Eq. (9.31)/(9.32) and changes the second Z_1^a in Eq. (9.44) to Z_2^a . SV0Errata

The book's assumptions for this Chapter 9 set are important: flat road, negligible suspension jacking, rigid tyres, small pitch θ , small roll ϕ , small roll/pitch rates, and the simplified small-yaw-rate terms used between Eq. (9.29) and Eq. (9.39). The book explicitly states the flat-road/negligible-jacking assumption before the 3D dynamics derivation, then assumes negligible unsprung mass at first, and later gives an optional unsprung-mass correction. 978-94-017-8533-4

Corrected Chapter 9 EOM and tyre vertical loads

Define

$$L = h - d$$

where h is the vehicle CG height above the road and d is the VIP/no-roll-center height. Also,

$$r = \dot{\psi}, \quad l = a_1 + a_2.$$

The Chapter 9 translational accelerations are

$$\begin{aligned} a_x &= (\dot{u} - vr) + L\ddot{\theta} \\ a_y &= (\dot{v} + ur) - L\dot{\phi}. \end{aligned}$$

Therefore,

$$X = ma_x, \quad Y = ma_y.$$

The corrected negligible-unsprung-mass EOM are

$$\begin{aligned} m \left[(\dot{u} - vr) + L\ddot{\theta} \right] &= X \\ m \left[(\dot{v} + ur) - L\dot{\phi} \right] &= Y \\ 0 &= Z \\ (J_x + mL^2) \ddot{\phi} - J_{xz} \dot{r} - mL(\dot{v} + ur) &= L_M \\ (J_y + mL^2) \ddot{\theta} + mL(\dot{u} - vr) &= M_M \\ J_z \dot{r} - J_{xz} \dot{\phi} + mL\dot{u} \phi &= N. \end{aligned}$$

These are exactly the structure of Eq. (9.39), after the upstream angular-momentum errata are applied. The book derives a_G , $\dot{K}_G^a$, and $MG \times ma_G$, then combines them into Eq. (9.39). 978-94-017-8533-4

The equivalent compact form is

$$\begin{aligned} J_x \ddot{\phi} - J_{xz} \dot{r} - ma_y L &= L_M \\ J_y \ddot{\theta} + ma_x L &= M_M \\ J_z \dot{r} - J_{xz} \dot{\phi} + ma_x L \phi &= N. \end{aligned}$$

For implementation, I would use the first Eq. (9.39) form, because it shows explicitly which acceleration terms were retained.

Corrected right-hand sides

The corrected resultant forces and moments are

$$\begin{aligned} X &= X_1 + X_2 - \frac{1}{2} \rho S C_x u^2 \\ Y &= Y_1 + Y_2 \\ Z &= Z_1 + Z_1^a + Z_2 + Z_2^a - mg. \end{aligned}$$

The last equation is one of the key corrections: the printed book repeats Z_1^a , but the errata says it must be Z_1^a and Z_2^a . SV0Errata

The corrected roll, pitch, and yaw moments are

$$\begin{aligned} L_M &= -k_\phi \phi - c_\phi \dot{\phi} + mgL\phi \\ M_M &= -k_\theta \theta - c_\theta \dot{\theta} \\ N &= (Y_1 a_1 - Y_2 a_2) + (\Delta X_1 t_1 + \Delta X_2 t_2). \end{aligned}$$

with

$$\begin{aligned} k_\phi &= k_{\phi 1} + k_{\phi 2}, & c_\phi &= c_{\phi 1} + c_{\phi 2} \\ k_\theta &= k_1 a_1^2 + k_2 a_2^2, & c_\theta &= c_1 a_1^2 + c_2 a_2^2. \end{aligned}$$

Fully substituted corrected EOM

Substituting the corrected right-hand sides gives the final corrected working equations:

$$\begin{aligned} m \left[(\dot{u} - vr) + L\ddot{\theta} \right] &= X_1 + X_2 - \frac{1}{2} \rho S C_x u^2 \\ m \left[(\dot{v} + ur) - L\dot{\phi} \right] &= Y_1 + Y_2 \\ Z_1 + Z_1^a + Z_2 + Z_2^a - mg &= 0 \\ (J_x + mL^2) \ddot{\phi} - J_{xz} \dot{r} - mL(\dot{v} + ur) &= -k_\phi \phi - c_\phi \dot{\phi} + mgL\phi \\ (J_y + mL^2) \ddot{\theta} + mL(\dot{u} - vr) &= -k_\theta \theta - c_\theta \dot{\theta} \\ J_z \dot{r} - J_{xz} \dot{\phi} + mL\dot{u} \phi &= (Y_1 a_1 - Y_2 a_2) + (\Delta X_1 t_1 + \Delta X_2 t_2). \end{aligned}$$

This is the version I would trust as the corrected Chapter 9 EOM set.

Corrected load-transfer equations

The lateral load transfers are

$$\begin{aligned} \Delta Z_1 t_1 &= Y_1 q_1 + k_{\phi 1} \phi + c_{\phi 1} \dot{\phi} \\ \Delta Z_2 t_2 &= Y_2 q_2 + k_{\phi 2} \phi + c_{\phi 2} \dot{\phi}. \end{aligned}$$

Therefore,

$$\begin{aligned} \Delta Z_1 &= \frac{Y_1 q_1 + k_{\phi 1} \phi + c_{\phi 1} \dot{\phi}}{t_1} \\ \Delta Z_2 &= \frac{Y_2 q_2 + k_{\phi 2} \phi + c_{\phi 2} \dot{\phi}}{t_2}. \end{aligned}$$

Adding the two axle equations gives

$$\Delta Z_1 t_1 + \Delta Z_2 t_2 = k_\phi \phi + c_\phi \dot{\phi} + Y_1 q_1 + Y_2 q_2.$$

Since the book defines

$$Yd = Y_1 q_1 + Y_2 q_2,$$

and $Y' = ma_y$,

$$\Delta Z_1 t_1 + \Delta Z_2 t_2 = k_\phi \phi + c_\phi \dot{\phi} + m a_y d.$$

Combining this with the roll EOM gives the total lateral load transfer

$$\Delta Z_1 t_1 + \Delta Z_2 t_2 = m a_y h + m g L \phi - \left(J_x \ddot{\phi} - J_{xz} \ddot{r} \right).$$

This matches the corrected Chapter 9 logic from Eq. (9.45) to Eq. (9.48). 978-94-017-8533-4

The corrected longitudinal load transfer is

$$\Delta Z = \frac{-Xh + J_y \ddot{\theta}}{l}.$$

Since $X = m a_x$,

$$\Delta Z = \frac{-m a_x h + J_y \ddot{\theta}}{l}.$$

This is Eq. (9.49), and the signs are correct: braking means $a_x < 0$, so $-m a_x h/l > 0$, increasing front axle load. 978-94-017-8533-4

Corrected individual tyre vertical loads

Chapter 3 defines

$$Z_1 = F_{z11} + F_{z12}, \quad Z_2 = F_{z21} + F_{z22}$$

and

$$\Delta Z_1 = \frac{F_{z12} - F_{z11}}{2}, \quad \Delta Z_2 = \frac{F_{z22} - F_{z21}}{2}.$$

Therefore,

$$\begin{aligned} F_{z11} &= Z_{11} = \frac{Z_1}{2} - \Delta Z_1 \\ F_{z12} &= Z_{12} = \frac{Z_1}{2} + \Delta Z_1 \\ F_{z21} &= Z_{21} = \frac{Z_2}{2} - \Delta Z_2 \\ F_{z22} &= Z_{22} = \frac{Z_2}{2} + \Delta Z_2. \end{aligned}$$

This sign convention means positive ΔZ_i increases wheel $i2$ and decreases wheel $i1$. 978-94-017-8533-4

The static axle loads are

$$\begin{aligned} Z_1^0 &= \frac{m g a_2}{l} \\ Z_2^0 &= \frac{m g a_1}{l}, \end{aligned}$$

The corrected dynamic axle loads are

$$\begin{aligned} Z_1 &= Z_1^0 + \Delta Z - Z_1^a \\ Z_2 &= Z_2^0 - \Delta Z - Z_2^a. \end{aligned}$$

Therefore the corrected individual tyre vertical loads are

$$\begin{aligned} Z_{11} &= \frac{1}{2} \left(Z_1^0 + \Delta Z - Z_1^a \right) - \Delta Z_1 \\ Z_{12} &= \frac{1}{2} \left(Z_1^0 + \Delta Z - Z_1^a \right) + \Delta Z_1 \\ Z_{21} &= \frac{1}{2} \left(Z_2^0 - \Delta Z - Z_2^a \right) - \Delta Z_2 \\ Z_{22} &= \frac{1}{2} \left(Z_2^0 - \Delta Z - Z_2^a \right) + \Delta Z_2. \end{aligned}$$

Substituting ΔZ , ΔZ_1 , and ΔZ_2 :

$$\begin{aligned} Z_{11} &= \frac{1}{2} \left[\frac{m g a_2}{l} + \frac{-m a_x h + J_y \ddot{\theta}}{l} - Z_1^a \right] - Y_1 q_1 + k_{\phi 1} \phi + c_{\phi 1} \dot{\phi} \\ Z_{12} &= \frac{1}{2} \left[\frac{m g a_2}{l} + \frac{-m a_x h + J_y \ddot{\theta}}{l} - Z_1^a \right] + Y_1 q_1 + k_{\phi 1} \phi + c_{\phi 1} \dot{\phi} \\ Z_{21} &= \frac{1}{2} \left[\frac{m g a_1}{l} - \frac{m a_x h + J_y \ddot{\theta}}{l} - Z_2^a \right] - Y_2 q_2 + k_{\phi 2} \phi + c_{\phi 2} \dot{\phi} \\ Z_{22} &= \frac{1}{2} \left[\frac{m g a_1}{l} - \frac{m a_x h + J_y \ddot{\theta}}{l} - Z_2^a \right] + Y_2 q_2 + k_{\phi 2} \phi + c_{\phi 2} \dot{\phi}, \end{aligned}$$

Equivalently, the rear loads can be written as

$$\begin{aligned} Z_{21} &= \frac{1}{2} \left[\frac{m g a_1}{l} + \frac{m a_x h - J_y \ddot{\theta}}{l} - Z_2^a \right] - Y_2 q_2 + k_{\phi 2} \phi + c_{\phi 2} \dot{\phi} \\ Z_{22} &= \frac{1}{2} \left[\frac{m g a_1}{l} + \frac{m a_x h - J_y \ddot{\theta}}{l} - Z_2^a \right] + Y_2 q_2 + k_{\phi 2} \phi + c_{\phi 2} \dot{\phi}. \end{aligned}$$

These are the tyre vertical loads I would use.

Sanity checks

The vertical-load sum is

$$Z_{11} + Z_{12} + Z_{21} + Z_{22} = m g - Z_1^a - Z_2^a.$$

This matches the corrected vertical force equation

$$Z_1 + Z_1^a + Z_2 + Z_2^a - m g = 0.$$

If $Z_i^a < 0$, meaning aerodynamic downforce under the book's sign convention, then $-Z_i^a > 0$, so the tyre normal loads increase.

For braking with no pitch acceleration,

$$a_x < 0, \quad \ddot{\theta} = 0,$$

so

$$\Delta Z = \frac{-m a_x h}{l} > 0,$$

which increases front load and decreases rear load. That is physically correct.

For positive lateral acceleration with positive ΔZ_i , the book's definition gives

$$Z_{i2} > Z_{i1},$$

so wheel $i2$ receives more vertical load than wheel $i1$. This is consistent with the Chapter 3 definition of ΔZ_i . 978-94-017-8533-4

So after this re-check, I would proceed with exactly the EOM and tyre-load equations above for the corrected Chapter 9 model.

已思考 3m 42s ›

Yes. Let's slow it down and rebuild the logic from the ground up.

I am fully confident in the **corrected** Chapter 9 equations as the book's intended linearized model, after applying the errata. They are **not full nonlinear 3D multibody equations**; they are the book's small-roll/small-pitch, flat-road, rigid-tyre, negligible-jacking handling-with-roll equations. The errata must be applied to Eq. (9.31), Eq. (9.32), and Eq. (9.44).

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Big picture: how Chapter 3 and Chapter 9 connect

Chapter 3 does this:

planar vehicle motion + tyre forces + algebraic tyre loads

So Chapter 3 has u, v, r , longitudinal/lateral tyre forces, yaw moment, and algebraic vertical-load equations. It defines the axle vertical loads Z_1, Z_2 , the lateral load transfers $\Delta Z_1, \Delta Z_2$, and the individual tyre loads Z_{ij} .

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Chapter 9 does **not throw Chapter 3 away**. Instead, it adds dynamic roll and pitch:

$$\phi(t), \dot{\phi}(t), \ddot{\phi}(t), \quad \theta(t), \dot{\theta}(t), \ddot{\theta}(t)$$

So Chapter 9 is basically:

Chapter 3 handling model + roll/pitch inertia + roll/pitch suspension stiffness and damping.

The book explicitly says that the Chapter 3 analysis is extended in Chapter 9, with rigid tyres assumed, and that the tyre equations themselves are still the Chapter 3 tyre equations.

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Step 1: Do not define u, v at G anymore

In Chapter 3, the velocity variables u, v are naturally tied to the vehicle body/CG planar motion.

In Chapter 9, because the body rolls, defining the lateral velocity at G becomes messy: G moves sideways just because the body rolls. So the book introduces the **vehicle invariant point**, VIP, called M . The point M is chosen so that it is a better point for defining vehicle position and lateral velocity when roll exists. The book assumes flat road and negligible suspension jacking here.

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Yes. The cleanest way is to treat Chapter 9 as **Chapter 3 plus dynamic roll and pitch**, not as a completely new vehicle model.

The book's own structure supports that reading: Chapter 9 first defines yaw–pitch–roll, then introduces the vehicle invariant point M , then derives the 3D rigid-body dynamics, and finally says the Chapter 3 tyre equations still apply; only load transfers are updated because roll/pitch angular accelerations now appear.

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I will use

$$H = h - d$$

where H is the vertical distance from the vehicle invariant point M to the CG G . I use H instead of L to avoid confusing it with roll moment L_M or wheelbase L .

0. The mental map

The model has three layers:

Kinematics ⇒ Rigid-body dynamics ⇒ Tyre forces and vertical loads

Chapter 3 mostly does this for a **planar vehicle**:

$$u, \, v, \, r$$

Chapter 9 adds:

$$\phi, \, \dot{\phi}, \, \ddot{\phi} \\ \theta, \, \dot{\theta}, \, \ddot{\theta}$$

So Chapter 9 is not a full 14DOF multibody model. It is a **linearized handling-with-roll/pitch** model.

Part A — What Chapter 3 already gave you

In Chapter 3, the vehicle is planar. There is no roll inertia and no pitch inertia. The velocity is written as

$$\mathbf{V}_G = u\mathbf{i} + v\mathbf{j}$$

so the acceleration of G is

$$\mathbf{a}_G = (\dot{u} - vr)\mathbf{i} + (\dot{v} + ur)\mathbf{j}.$$

Define

$$a_x = \dot{u} - vr \\ a_y = \dot{v} + ur.$$

Then force balance gives

$$ma_x = X \\ ma_y = Y.$$

Yaw balance gives

$$J_x \dot{r} = N.$$

The tyre/road forces are collected as

$$X = X_1 + X_2 - X_a \\ Y = Y_1 + Y_2 \\ N = Y_1a_1 - Y_2a_2 + \Delta X_1t_1 + \Delta X_2t_2.$$

That is the basic Chapter 3 planar handling model. Chapter 3 also defines axle vertical loads Z_1, Z_2 , lateral load transfers $\Delta Z_1, \Delta Z_2$, and individual tyre vertical loads Z_{ij} . Specifically, it defines $Z_1 = F_{z11} + F_{z12}$, $Z_2 = F_{z21} + F_{z22}$, $\Delta Z_1 = (F_{z12} - F_{z11})/2$, and $\Delta Z_2 = (F_{z22} - F_{z21})/2$. Inverting those definitions gives $F_{z11} = Z_1/2 - \Delta Z_1$, $F_{z12} = Z_1/2 + \Delta Z_1$, and similarly for the rear axle. 978-94-017-8331-4

The important Chapter 3 idea is this:

Equilibrium gives total vertical load and total load transfer,

but

suspension geometry/stiffness decides how that load transfer is split front/rear.

That is why Chapter 3 has both global equilibrium equations and suspension equations.

Part B — Why Chapter 9 introduces the VIP $\bar{M}$

Once the vehicle rolls, the CG G moves sideways just because the body rolls. If we define lateral velocity at G , the lateral velocity v becomes contaminated by roll motion.

So Chapter 9 defines a better point: the **vehicle invariant point**, $\bar{M}$. The book defines $\bar{M}$ as the body point that, in the reference configuration, coincides with the no-roll center $\bar{Q}$. Then the vehicle velocity is defined at $\bar{M}$, not at G . 978-94-017-8331-4

So Chapter 9 defines

$$\mathbf{V}_{\bar{M}} = u\mathbf{i}_1 + v\mathbf{j}_1.$$

Then, because $\mathbf{i}_1, \mathbf{j}_1$ rotate with yaw rate $r = \dot{\psi}$,

$$\mathbf{a}_{\bar{M}} = (\dot{u} - vr)\mathbf{i}_1 + (\dot{v} + ur)\mathbf{j}_1.$$

This is still the same-looking formula as Chapter 3, but now it is the acceleration of $\bar{M}$, not G .

That is the first key idea:

In Chapter 9, u, v belong to $\bar{M}$, not G .

Part C — Yaw, pitch, roll kinematics

Chapter 9 uses the yaw-pitch-roll sequence

$$S_0 \overset{\psi}{\rightarrow} S_1 \overset{\theta}{\rightarrow} S_2 \overset{\phi}{\rightarrow} S_3.$$

The small-angle result for the body-fixed axes is

$$\mathbf{i}_3 \simeq \mathbf{i}_1 - \theta \mathbf{k}_1 \\ \mathbf{j}_3 \simeq \mathbf{j}_1 + \phi \mathbf{k}_1 \\ \mathbf{k}_3 \simeq \theta \mathbf{i}_1 - \phi \mathbf{j}_1 + \mathbf{k}_1.$$

The angular velocity components are approximately

$$p \simeq \dot{\phi} - \dot{\psi} \theta \\ q \simeq \dot{\theta} + \dot{\psi} \phi \\ r \simeq \dot{\psi}.$$

Then, for the simplified Chapter 9 final equations, the book further discards small yaw-rate/yaw-acceleration products, so the working approximation is essentially

$$p \simeq \dot{\phi}, \qquad q \simeq \dot{\theta}, \qquad r \simeq \dot{\psi}.$$

This approximation step is very important. Chapter 9 is not keeping every nonlinear term.

Part D — Position of G relative to $\bar{M}$

The CG is above the VIP by distance

$$H = h - d.$$

So

$$\mathbf{MG} = H\mathbf{k}_3.$$

Using the small-angle expression for $\mathbf{k}_3$,

$$\mathbf{MG} \simeq H(\theta \mathbf{i}_1 - \phi \mathbf{j}_1 + \mathbf{k}_1).$$

This equation is the geometric bridge between $\bar{M}$ and G .

It says:

pitch moves G longitudinally by $H\theta$,

roll moves G laterally by $-H\phi$.

Part E — Derive acceleration of G

Rigid-body acceleration formula:

$$\mathbf{a}_G = \mathbf{a}_{\bar{M}} + \dot{\boldsymbol{\Omega}} \times \mathbf{MG} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{MG}).$$

Chapter 9 simplifies this. After discarding the small yaw-rate/yaw-acceleration terms, the result is

$$\mathbf{a}_G \simeq (\dot{u} - vr)\mathbf{i}_1 + (\dot{v} + ur)\mathbf{j}_1 + H\ddot{\theta}\mathbf{i}_1 - H\ddot{\phi}\mathbf{j}_1.$$

Collect terms:

$$\mathbf{a}_G \simeq \left[(\dot{u} - vr) + H\ddot{\theta} \right] \mathbf{i}_1 + \left[(\dot{v} + ur) - H\ddot{\phi} \right] \mathbf{j}_1.$$

So Chapter 9 defines the effective accelerations

$$a_x = (\dot{u} - vr) + H\ddot{\theta} \\ a_y = (\dot{v} + ur) - H\ddot{\phi}.$$

This is the second key idea:

Pitch acceleration modifies longitudinal acceleration of G .

Roll acceleration modifies lateral acceleration of G .

That is why Chapter 9 force equations become different from Chapter 3.

Part F — Translational EOM

Now use

$$m\mathbf{a}_G = \mathbf{F}.$$

The external resultant is

$$\mathbf{F} = X\mathbf{i}_1 + Y\mathbf{j}_1 + Z\mathbf{k}_1.$$

Therefore,

$$m \left[(\dot{u} - vr) + H\ddot{\theta} \right] = X \\ m \left[(\dot{v} + ur) - H\ddot{\phi} \right] = Y \\ 0 = Z.$$

These are the first three Chapter 9 EOM.

Compare with Chapter 3:

$$m(\dot{u} - vr) = X \\ m(\dot{v} + ur) = Y.$$

So the only change is:

$$\dot{u} - vr \quad \rightarrow \quad \dot{u} - vr + H\ddot{\theta}$$

$$\dot{v} + ur \quad \rightarrow \quad \dot{v} + ur - H\ddot{\phi}.$$

Part G — Angular momentum derivative

The angular momentum equation is the part that makes Chapter 9 look scary.

The book first writes the angular momentum derivative in the body-fixed frame S_b . The printed Eq. (9.31) and Eq. (9.32) have sign errors, and the errata corrects them. The errata says the signs inside the last parentheses of Eq. (9.31) and the first row of Eq. (9.32) must be corrected; it also corrects Eq. (9.44)'s repeated aerodynamic vertical load term. SVDErrata

After applying the errata, assuming

$$J_{xy} \simeq 0, \qquad J_{yz} \simeq 0,$$

and then applying the same Chapter 9 small-term approximations, the angular momentum derivative becomes

$$\dot{\mathbf{K}}_G^r \simeq (J_x\ddot{\phi} - J_{xx}\dot{r})\mathbf{i}_1 + J_y\ddot{\theta}\mathbf{j}_1 + (J_z\dot{r} - J_{zx}\ddot{\phi})\mathbf{k}_1.$$

This gives three inertial angular terms:

Roll direction:

$$J_x\ddot{\phi} - J_{xx}\dot{r}$$

Pitch direction:

$$J_y\ddot{\theta}$$

Yaw direction:

$$J_z\dot{r} - J_{zx}\ddot{\phi}.$$

The product of inertia J_{zx} couples roll and yaw.

Part H — Why the torque equation is about $\bar{M}$, not G

The standard angular momentum equation about G is

$$\dot{\mathbf{K}}_G^r = \mathbf{M}_G.$$

But Chapter 9 wants moments about M . For a moving point M , the torque equation becomes

$$\dot{\mathbf{K}}_G^r + \mathbf{M}\mathbf{G} \times m\mathbf{a}_G = \mathbf{M}_M.$$

So you need one extra term:

$$\mathbf{M}\mathbf{G} \times m\mathbf{a}_G.$$

Using the same simplified Chapter 9 expressions,

$$\mathbf{M}\mathbf{G} \times m\mathbf{a}_G \simeq m \left[(H^2\ddot{\phi} - H(\dot{v} + ur))\mathbf{i}_1 + (H^2\ddot{\theta} + H(\dot{u} - vr))\mathbf{j}_1 + H\dot{u}\phi\mathbf{k}_1 \right].$$

This term explains the mH^2 terms in the final EOM.

For example, the roll-direction part is

$$mH^2\ddot{\phi} - mH(\dot{v} + ur).$$

Add that to the angular momentum roll part:

$$J_x\ddot{\phi} - J_{xx}\dot{r}.$$

Then the total roll left-hand side becomes

$$(J_x + mH^2)\ddot{\phi} - J_{xx}\dot{r} - mH(\dot{v} + ur).$$

That is exactly the roll equation.

Part I — Rotational EOM

Now combine

$$\dot{\mathbf{K}}_G^r + \mathbf{M}\mathbf{G} \times m\mathbf{a}_G = \mathbf{M}_M.$$

Roll equation:

$$(J_x + mH^2)\ddot{\phi} - J_{xx}\dot{r} - mH(\dot{v} + ur) = L_M.$$

Pitch equation:

$$(J_y + mH^2)\ddot{\theta} + mH(\dot{u} - vr) = M_M.$$

Yaw equation:

$$J_z\dot{r} - J_{xx}\ddot{\phi} + mH\dot{u}\phi = N.$$

So the full left-hand-side EOM are:

$$m \left[(\dot{u} - vr) + H\ddot{\theta} \right] = X$$

$$m \left[(\dot{v} + ur) - H\ddot{\phi} \right] = Y$$

$$0 = Z$$

$$(J_x + mH^2)\ddot{\phi} - J_{xx}\dot{r} - mH(\dot{v} + ur) = L_M$$

$$(J_y + mH^2)\ddot{\theta} + mH(\dot{u} - vr) = M_M$$

$$J_z\dot{r} - J_{xx}\ddot{\phi} + mH\dot{u}\phi = N.$$

That is Chapter 9's corrected dynamic core.

Part J — Right-hand sides of the EOM

Now we insert external forces and moments.

The force resultants are

$$X = X_1 + X_2 - \frac{1}{2}\rho SC_x u^2$$

$$Y = Y_1 + Y_2$$

$$Z = Z_1 + Z_1^a + Z_2 + Z_2^a - mg.$$

The last one is corrected by the errata. The printed book repeats Z_1^a , but the correction is

$$Z = Z_1 + Z_1^a + Z_2 + Z_2^a - mg.$$

SVDErrata

The moment resultants are

$$L_M = -k_\phi\phi - c_\phi\dot{\phi} + mgH\phi$$

$$M_M = -k_\theta\theta - c_\theta\dot{\theta}$$

$$N = (Y_1a_1 - Y_2a_2) + (\Delta X_1t_1 + \Delta X_2t_2).$$

The roll stiffness/damping are

$$k_\phi = k_{\phi 1} + k_{\phi 2}$$

$$c_\phi = c_{\phi 1} + c_{\phi 2}.$$

The pitch stiffness/damping come from Chapter 8's ride/pitch model:

$$k_\theta = k_1a_1^2 + k_2a_2^2$$

$$c_\theta = c_1a_1^2 + c_2a_2^2.$$

Chapter 8 derives the pitch equation using front/rear suspension forces and obtains the stiffness and damping matrix entries containing $k_1a_1^2 + k_2a_2^2$ and $c_1a_1^2 + c_2a_2^2$. SVDErrata

Part K — Fully substituted corrected Chapter 9 EOM

Substitute the right-hand sides:

$$m \left[(\dot{u} - vr) + H\ddot{\theta} \right] = X_1 + X_2 - \frac{1}{2}\rho SC_x u^2$$

$$m \left[(\dot{v} + ur) - H\ddot{\phi} \right] = Y_1 + Y_2$$

$$Z_1 + Z_1^a + Z_2 + Z_2^a - mg = 0$$

$$(J_x + mH^2)\ddot{\phi} - J_{xx}\dot{r} - mH(\dot{v} + ur) = -k_\phi\phi - c_\phi\dot{\phi} + mgH\phi$$

$$(J_y + mH^2)\ddot{\theta} + mH(\dot{u} - vr) = -k_\theta\theta - c_\theta\dot{\theta}$$

$$J_z\dot{r} - J_{xx}\ddot{\phi} + mH\dot{u}\phi = (Y_1a_1 - Y_2a_2) + (\Delta X_1t_1 + \Delta X_2t_2).$$

These are the EOM I would implement for the corrected Chapter 9 model.

Part L — Now derive tyre vertical loads

This is where Chapter 3 and Chapter 9 combine.

L1. Start from Chapter 3 definitions

For the front axle:

$$Z_1 = Z_{11} + Z_{12}$$

$$\Delta Z_1 = \frac{Z_{12} - Z_{11}}{2}.$$

Therefore,

$$Z_{11} = \frac{Z_1}{2} - \Delta Z_1$$

$$Z_{12} = \frac{Z_1}{2} + \Delta Z_1.$$

For the rear axle:

$$Z_2 = Z_{21} + Z_{22}$$

$$\Delta Z_2 = \frac{Z_{22} - Z_{21}}{2}.$$

Therefore,

$$Z_{21} = \frac{Z_2}{2} - \Delta Z_2$$

$$Z_{22} = \frac{Z_2}{2} + \Delta Z_2.$$

These are just definitions. No dynamics yet.

L2. Static axle loads

The static front and rear axle loads are

$$\begin{aligned}Z_1^0 &= \frac{mga_2}{l} \\ Z_2^0 &= \frac{mga_1}{l}.\end{aligned}$$

where

$$l = a_1 + a_2.$$

L3. Longitudinal load transfer

Chapter 9 updates the longitudinal load transfer because pitch acceleration now matters:

$$\Delta Z = \frac{-Xh + J_y\ddot{\theta}}{l}.$$

Since

$$X = ma_x,$$

and

$$a_x = (\dot{u} - vr) + H\ddot{\theta},$$

you may also write

$$\Delta Z = \frac{-ma_xh + J_y\ddot{\theta}}{l}.$$

Then the axle loads are

$$\begin{aligned}Z_1 &= Z_1^0 + \Delta Z - Z_1^a \\ Z_2 &= Z_2^0 - \Delta Z - Z_2^a.\end{aligned}$$

This sign convention is physically sensible:

During braking,

$$a_x < 0,$$

so

$$-ma_xh > 0,$$

therefore

$$\Delta Z > 0,$$

so the front axle load increases.

L4. Lateral load transfer at each axle

For rigid tyres, Chapter 3 gives

$$\Delta Z_i t_i = Y_i q_i + k_{\phi i} \phi.$$

Chapter 9 adds damper contribution:

$$\Delta Z_i t_i = Y_i q_i + k_{\phi i} \phi + c_{\phi i} \dot{\phi}.$$

So:

Front axle:

$$\Delta Z_1 t_1 = Y_1 q_1 + k_{\phi 1} \phi + c_{\phi 1} \dot{\phi}.$$

Rear axle:

$$\Delta Z_2 t_2 = Y_2 q_2 + k_{\phi 2} \phi + c_{\phi 2} \dot{\phi}.$$

Therefore,

$$\begin{aligned}\Delta Z_1 &= \frac{Y_1 q_1 + k_{\phi 1} \phi + c_{\phi 1} \dot{\phi}}{t_1} \\ \Delta Z_2 &= \frac{Y_2 q_2 + k_{\phi 2} \phi + c_{\phi 2} \dot{\phi}}{t_2}.\end{aligned}$$

Interpretation:

$$Y_i q_i$$

is the direct geometric/no-roll-center contribution.

$$k_{\phi i} \phi$$

is the elastic roll-stiffness contribution.

$$c_{\phi i} \dot{\phi}$$

is the damper contribution added in Chapter 9.

L5. Individual tyre vertical loads

Now plug everything into the Chapter 3 definitions.

Front-left/front-right:

$$\begin{aligned}Z_{11} &= \frac{1}{2} \left(Z_1^0 + \Delta Z - Z_1^a \right) - \Delta Z_1 \\ Z_{12} &= \frac{1}{2} \left(Z_1^0 + \Delta Z - Z_1^a \right) + \Delta Z_1.\end{aligned}$$

Rear-left/rear-right:

$$\begin{aligned}Z_{21} &= \frac{1}{2} \left(Z_2^0 - \Delta Z - Z_2^a \right) - \Delta Z_2 \\ Z_{22} &= \frac{1}{2} \left(Z_2^0 - \Delta Z - Z_2^a \right) + \Delta Z_2.\end{aligned}$$

Substitute the explicit formulas:

$$\begin{aligned}Z_{11} &= \frac{1}{2} \left[\frac{mga_2}{l} + \frac{-ma_xh + J_y\ddot{\theta}}{l} - Z_1^a \right] - \frac{Y_1 q_1 + k_{\phi 1} \phi + c_{\phi 1} \dot{\phi}}{t_1} \\ Z_{12} &= \frac{1}{2} \left[\frac{mga_2}{l} + \frac{-ma_xh + J_y\ddot{\theta}}{l} - Z_1^a \right] + \frac{Y_1 q_1 + k_{\phi 1} \phi + c_{\phi 1} \dot{\phi}}{t_1} \\ Z_{21} &= \frac{1}{2} \left[\frac{mga_1}{l} - \frac{ma_xh + J_y\ddot{\theta}}{l} - Z_2^a \right] - \frac{Y_2 q_2 + k_{\phi 2} \phi + c_{\phi 2} \dot{\phi}}{t_2} \\ Z_{22} &= \frac{1}{2} \left[\frac{mga_1}{l} - \frac{ma_xh + J_y\ddot{\theta}}{l} - Z_2^a \right] + \frac{Y_2 q_2 + k_{\phi 2} \phi + c_{\phi 2} \dot{\phi}}{t_2}.\end{aligned}$$

These are the tyre vertical loads.

Part M — Important consistency checks

Check 1: total vertical force

Add all four tyre loads:

$$Z_{11} + Z_{12} + Z_{21} + Z_{22} = Z_1 + Z_2.$$

Using

$$\begin{aligned}Z_1 &= Z_1^0 + \Delta Z - Z_1^a \\ Z_2 &= Z_2^0 - \Delta Z - Z_2^a,\end{aligned}$$

we get

$$Z_1 + Z_2 = Z_1^0 + Z_2^0 - Z_1^a - Z_2^a.$$

Since

$$Z_1^0 + Z_2^0 = mg,$$

then

$$Z_{11} + Z_{12} + Z_{21} + Z_{22} = mg - Z_1^a - Z_2^a.$$

This matches the corrected vertical equilibrium equation:

$$Z_1 + Z_1^a + Z_2 + Z_2^a - mg = 0.$$

Good.

Check 2: braking load transfer

If braking and no pitch acceleration,

$$a_x < 0, \quad \ddot{\theta} = 0,$$

then

$$\Delta Z = \frac{-ma_xh}{l} > 0.$$

So

$$Z_1 = Z_1^0 + \Delta Z - Z_1^a$$

increases, and

$$Z_2 = Z_2^0 - \Delta Z - Z_2^a$$

decreases.

Good.

Check 3: lateral load-transfer sum

Add the front and rear lateral load transfer equations:

$$\Delta Z_1 t_1 + \Delta Z_2 t_2 = Y_1 q_1 + Y_2 q_2 + k_{\phi} \phi + c_{\phi} \dot{\phi}.$$

Chapter 9 defines

$$Yd = Y_1 q_1 + Y_2 q_2.$$

So

$$\Delta Z_1 t_1 + \Delta Z_2 t_2 = Y d + k_\phi \phi + c_\phi \dot{\phi}.$$

Using the roll EOM, this becomes

$$\Delta Z_1 t_1 + \Delta Z_2 t_2 = m a_y h + m g H \phi - (J_x \ddot{\phi} - J_{xz} \ddot{r}).$$

This is Chapter 9’s global lateral load-transfer result. It is a check, not a replacement for the front/rear split equations. Chapter 9 explicitly states that the main difference from the Chapter 3 model is that load transfers now depend on body angular accelerations. 978-94-017-8533-4

The shortest possible summary

Chapter 3:

$$m(\dot{u} - vr) = X, \quad m(\dot{v} + ur) = Y, \quad J_z \dot{r} = N$$

Chapter 9 changes the accelerations because u, v are defined at M , while dynamics needs G :

$$a_x = (\dot{u} - vr) + H \ddot{\theta}$$

$$a_y = (\dot{v} + ur) - H \ddot{\phi}$$

Then Chapter 9 adds roll/pitch/yaw rotational dynamics about M :

$$(J_x + m H^2) \ddot{\phi} - J_{xz} \dot{r} - m H (\dot{v} + ur) = L_M$$

$$(J_y + m H^2) \ddot{\theta} + m H (\dot{u} - vr) = M_M$$

$$J_z \dot{r} - J_{zx} \ddot{\phi} + m H \dot{u} \phi = N$$

The tyre vertical loads are still Chapter 3 style:

$$Z_{i1} = \frac{Z_i}{2} - \Delta Z_{i1}, \quad Z_{i2} = \frac{Z_i}{2} + \Delta Z_{i1}$$

but Chapter 9 updates

$$\Delta Z = \frac{-m a_x h + J_y \ddot{\theta}}{l}$$

and

$$\Delta Z_i t_i = Y_i q_i + k_{\phi i} \phi + c_{\phi i} \dot{\phi}.$$

That is the whole logic. Chapter 9 is confusing because it mixes three things at once: changing the reference point from G to M , adding roll/pitch inertial terms, and updating the Chapter 3 load-transfer formulas.

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SVDerrata(1).pdf
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I want you to provide me a new learner instruction as detailed as possible for the derivations of EOMs of Chapter 3 as well as Chapter 9 together with the equations of tyre loads. This instruction should be used by me companying with the book reading, so that I can best understand the derivations. Make sure you explain as detailed as possible with everything in the book and the correction document, do not invent anything outside the book and the correction document, make sure your derivation and instructions are rigorous.

已思考 3m 46s ›

Below is a learner instruction you can read side-by-side with Chapters 3 and 9. I will keep the logic strictly inside the book plus the errata. I will use $H = h - d$ in Chapter 9 only to make the equations easier to read; in the book this same quantity is written as $h - d$.

The most important warning before reading: do not read the printed equations blindly. The errata corrects several Chapter 3 equations, especially tire-slip signs, aerodynamic-load notation, Eq. (3.110), and Eq. (3.122), and it corrects Chapter 9 Eq. (9.31), Eq. (9.32), and Eq. (9.44). SVDerrata SVDerrata

0. The correct mental map

The book builds the vehicle model from three kinds of equations:

congruence / kinematic equations

equilibrium / dynamic equations

constitutive / tyre equations

Chapter 3 explicitly says the vehicle model is made of these three sets. The assumptions there are flat road, negligible suspension deflections, negligible tyre vertical deformations, small steering angles, and a rigid steering system; under these assumptions the vehicle body has planar motion and the wheel centers have fixed positions relative to the body. 978-94-017-8533-4

So when you read the derivation, separate the model into these layers:

vehicle motion $\rightarrow$ wheel velocities/slips $\rightarrow$ tyre forces $\rightarrow$ vehicle equilibrium $\rightarrow$ vertical tyre loads.

Chapter 9 keeps the same structure, but adds dynamic roll and pitch.

1. Chapter 3: planar vehicle model

1.1 Start from the Chapter 3 assumptions

Chapter 3 assumes the road is flat and the vehicle remains in its reference configuration except for steering angles. So a_1, a_2, l, t_1, t_2, h are constant. The vehicle body has planar motion parallel to the road. 978-94-017-8533-4

That means the main vehicle motion variables are

$$u, \quad v, \quad r.$$

Here

u = longitudinal velocity of G ,

v = lateral velocity of G ,

$r = \dot{\psi}$ = yaw rate.

The acceleration of G is

$$\mathbf{a}_G = (\dot{u} - vr)\mathbf{i} + (\dot{v} + ur)\mathbf{j}.$$

Therefore define

$$a_x = \dot{u} - vr,$$

$$a_y = \dot{v} + ur.$$

This is the first key idea in Chapter 3:

The terms $-vr$ and ur appear because the vehicle frame rotates with yaw rate r .

1.2 Road-tire tangential forces: convert wheel-frame forces into vehicle-frame forces

For each tyre, the book first has wheel-frame forces F_{xij}, F_{yij} . Because a wheel can be steered by δ_{ij} , the vehicle-frame components are

$$X_{ij} = F_{xij} \cos \delta_{ij} - F_{yij} \sin \delta_{ij},$$

$$Y_{ij} = F_{xij} \sin \delta_{ij} + F_{yij} \cos \delta_{ij}.$$

Then the axle sums are

$$X_1 = X_{11} + X_{12}, \quad X_2 = X_{21} + X_{22},$$

$$Y_1 = Y_{11} + Y_{12}, \quad Y_2 = Y_{21} + Y_{22}.$$

The left-right longitudinal and lateral force differences are

$$\Delta X_1 = \frac{X_{12} - X_{11}}{2}, \quad \Delta X_2 = \frac{X_{22} - X_{21}}{2},$$

$$\Delta Y_1 = \frac{Y_{12} - Y_{11}}{2}, \quad \Delta Y_2 = \frac{Y_{22} - Y_{21}}{2}.$$

These definitions are given in the Chapter 3 force decomposition. 978-94-017-8533-4

1.3 First derive only the planar EOM

Use Newton-Euler equations:

$$m \mathbf{a}_G = \mathbf{F},$$

$$\dot{\mathbf{K}}'_G = M_G.$$

Because Chapter 3 is planar, the in-plane dynamic equations are

$$m a_x = X,$$

$$m a_y = Y,$$

$$J_z \dot{r} = N.$$

Then insert the force and yaw-moment definitions:

$$X = X_1 + X_2 - X_a,$$

$$Y = Y_1 + Y_2,$$

$$N = Y_1 a_1 - Y_2 a_2 + \Delta X_1 t_1 + \Delta X_2 t_2.$$

So the Chapter 3 planar EOM are

$$m(\dot{u} - vr) = X_1 + X_2 - X_a,$$

$$m(\dot{v} + ur) = Y_1 + Y_2,$$

$$J_z \dot{r} = Y_1 a_1 - Y_2 a_2 + \Delta X_1 t_1 + \Delta X_2 t_2.$$

In the final collected model, the book writes the drag as

$$X_a = \frac{1}{2} \rho S C_x u^2.$$

So the final practical form is

$$m(\dot{u} - vr) = X_1 + X_2 - \frac{1}{2} \rho S C_x u^2,$$

$$m(\dot{v} + ur) = Y_1 + Y_2,$$

$$J_z \dot{r} = Y_1 a_1 - Y_2 a_2 + \Delta X_1 t_1 + \Delta X_2 t_2.$$

This is Eq. (3.64) / Eq. (3.64').

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1.4 Now derive the vertical-force equations

Chapter 3 does not treat the vertical loads as independent inputs. They are constraint forces needed to keep the vehicle compatible with the flat road.

The book defines axle vertical loads as

$$Z_1 = F_{z11} + F_{z12},$$

$$Z_2 = F_{z21} + F_{z22}.$$

It defines lateral load transfers as

$$\Delta Z_1 = \frac{F_{z12} - F_{z11}}{2},$$

$$\Delta Z_2 = \frac{F_{z22} - F_{z21}}{2}.$$

Invert these definitions:

$$F_{z11} = Z_{11} = \frac{Z_1}{2} - \Delta Z_1,$$

$$F_{z12} = Z_{12} = \frac{Z_1}{2} + \Delta Z_1,$$

$$F_{z21} = Z_{21} = \frac{Z_2}{2} - \Delta Z_2,$$

$$F_{z22} = Z_{22} = \frac{Z_2}{2} + \Delta Z_2.$$

This is Eq. (3.60)–(3.62).

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Now use the out-of-plane equilibrium equations. In corrected notation, the vertical force equation is

$$Z_1 + Z_1^a + Z_2 + Z_2^a = mg.$$

The errata is important here: in Chapter 3 Eq. (3.65)/(3.66), the aerodynamic-load notation must be Z_1^a , Z_2^a , not repeated Z_1^t . SVDerrata

The pitching moment balance gives the longitudinal load transfer. The book obtains

$$\Delta Z = - \frac{ma_x h}{l} + \frac{J_{xx} r^3}{l}.$$

Equivalently,

$$\Delta Z = \frac{-ma_x h + J_{xx} r^3}{l}.$$

The static axle loads are

$$Z_1^0 = \frac{mga_2}{l},$$

$$Z_2^0 = \frac{mga_1}{l}.$$

Therefore the dynamic axle loads are

$$Z_1 = Z_1^0 + \Delta Z - Z_1^a,$$

$$Z_2 = Z_2^0 - \Delta Z - Z_2^a.$$

This is Eq. (3.73)–(3.75).

978-94-017-8533-4

1.5 Chapter 3 tyre vertical loads before suspension analysis

Substitute Z_1, Z_2 into the individual-load definitions:

$$Z_{11} = \frac{1}{2} \left(Z_1^0 + \Delta Z - Z_1^a \right) - \Delta Z_1,$$

$$Z_{12} = \frac{1}{2} \left(Z_1^0 + \Delta Z - Z_1^a \right) + \Delta Z_1,$$

$$Z_{21} = \frac{1}{2} \left(Z_2^0 - \Delta Z - Z_2^a \right) - \Delta Z_2,$$

$$Z_{22} = \frac{1}{2} \left(Z_2^0 - \Delta Z - Z_2^a \right) + \Delta Z_2.$$

Using $Z_1^t = \frac{1}{2} \rho_a S_a C_{z1} u^2$, $Z_2^a = \frac{1}{2} \rho_a S_a C_{z2} u^2$, and $\Delta Z = (-ma_x h + J_{xx} r^3)/l$, this becomes exactly the Chapter 3 Eq. (3.79) structure:

$$Z_{11} = \frac{1}{2} \left[\frac{mga_2}{l} - \frac{1}{2} \rho_a S_a C_{z1} u^2 - \frac{ma_x h - J_{xx} r^3}{l} \right] - \Delta Z_1,$$

$$Z_{12} = \frac{1}{2} \left[\frac{mga_2}{l} - \frac{1}{2} \rho_a S_a C_{z1} u^2 - \frac{ma_x h - J_{xx} r^3}{l} \right] + \Delta Z_1,$$

$$Z_{21} = \frac{1}{2} \left[\frac{mga_1}{l} - \frac{1}{2} \rho_a S_a C_{z2} u^2 + \frac{ma_x h - J_{xx} r^3}{l} \right] - \Delta Z_2,$$

$$Z_{22} = \frac{1}{2} \left[\frac{mga_1}{l} - \frac{1}{2} \rho_a S_a C_{z2} u^2 + \frac{ma_x h - J_{xx} r^3}{l} \right] + \Delta Z_2.$$

At this point you do not yet know $\Delta Z_1, \Delta Z_2$. Chapter 3 explicitly says these will be obtained from the suspension analysis. 978-94-017-8533-4

1.6 Why suspension analysis is needed

The global equilibrium equations only give the total lateral load transfer

$$\Delta Z_1 t_1 + \Delta Z_2 t_2.$$

They do not by themselves tell you how much goes to the front axle and how much goes to the rear axle.

That split depends on suspension geometry and roll stiffness. This is why Chapter 3 spends a long time on no-roll centers, suspension roll stiffnesses, tyre vertical stiffnesses, and suspension internal equilibrium.

The important suspension equations are:

$$\phi = \phi_i^e + \phi_i^p,$$

where ϕ_i^e is the suspension part and ϕ_i^p is the tyre-deflection part.

The book gives

$$\Delta Z_i t_i = k_{\phi i}^p \phi_i^p,$$

and

$$\Delta Z_i t_i = Y_i q_i + k_{\phi i}^e \phi_i^e.$$

The term $Y_i q_i$ is the geometric/no-roll-center contribution. The term $k_{\phi i}^e \phi_i^e$ is the elastic suspension contribution. These equations are collected in Sect. 3.8.11. 978-94-017-8533-4

1.7 Full Chapter 3 lateral load transfers with tyre vertical compliance

If tyre vertical compliance is included, Chapter 3 gives the full expressions Eq. (3.113)/(3.114). In compact learning form, remember the route:

$$\Delta Z_i t_i = k_{\phi i}^p \phi_i^p,$$

$$\Delta Z_i t_i = Y_i q_i + k_{\phi i}^e \phi_i^e,$$

$$\phi = \phi_i^e + \phi_i^p,$$

then solve the four-equation system for $\phi_1^e, \phi_2^e, \phi_1^p, \phi_2^p$, and then compute ΔZ_i .

The book's explicit expressions are long. The important reading instruction is: do not try to memorize Eq. (3.113)/(3.114); understand that they are just the algebraic solution of the suspension internal-equilibrium system. The book states that $\Delta Z_1, \Delta Z_2$ are linear functions of Y_1, Y_2 , namely

$$\Delta Z_1 = \xi_{11} Y_1 + \xi_{12} Y_2,$$

$$\Delta Z_2 = \xi_{21} Y_1 + \xi_{22} Y_2.$$

This is Eq. (3.115).

978-94-017-8533-4

Also, if you use the printed Eq. (3.110), apply the errata. The corrected suspension roll-angle expressions are

$$\phi_1^e = \frac{1}{k_{\phi 1}^e} \frac{k_{\phi 1} k_{\phi 2}}{k_{\phi}} \left[\frac{Y(h - q_0)}{k_{\phi 2}} - \frac{Y_1 q_1}{k_{\phi 1}^p} + \frac{Y_2 q_2}{k_{\phi 2}^p} \right],$$

$$\phi_2^e = \frac{1}{k_{\phi 2}^e} \frac{k_{\phi 1} k_{\phi 2}}{k_{\phi}} \left[\frac{Y(h - q_0)}{k_{\phi 1}} - \frac{Y_2 q_2}{k_{\phi 2}^p} + \frac{Y_1 q_1}{k_{\phi 1}^p} \right].$$

The errata says the same correction applies to the repeated Eq. (3.110) on page 88. SVDerrata

1.8 Chapter 3 rigid-tyre simplification

Chapter 9 later assumes rigid tyres. So this Chapter 3 simplification is especially important.

If tyre vertical deflections are neglected,

$$k_{\phi i}^p \rightarrow \infty,$$

and the total roll stiffness becomes the suspension roll stiffness contribution. Chapter 3 then gives

$$k_{\phi} \phi = Y(h - q_0),$$

and the lateral load transfers simplify to

$$\Delta Z_i t_i = k_{\phi i} \phi + Y_i q_i.$$

So for front and rear:

$$\Delta Z_1 t_1 = k_{\phi 1} \phi + Y_1 q_1,$$

$$\Delta Z_2 t_2 = k_{\phi 2} \phi + Y_2 q_2.$$

This is Eq. (3.117)/(3.118).

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This is the cleanest way to understand Chapter 9 later:

Chapter 9 starts from the rigid-tyre Chapter 3 load-transfer form, then adds damping and roll inertia.

1.9 Chapter 3 tyre constitutive closure

After the EOM and loads, the tyre model closes the system

$$\begin{aligned}F_{xij} &= F_{xij}(F_{xij}, \gamma_{ij}, \sigma_{xij}, \sigma_{yij}, \varphi_{ij}), \\F_{yij} &= F_{yij}(F_{xij}, \gamma_{ij}, \sigma_{xij}, \sigma_{yij}, \varphi_{ij}).\end{aligned}$$

So the vertical loads $Z_{ij} = F_{zij}$ feed directly into the tyre forces. The book states this in Sect. 3.11.2.

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The errata also corrects the friction feasibility condition Eq. (3.122):

$$X_{ij}^2 + Y_{ij}^2 \leq \mu_p Z_{ij}, \quad Z_{ij} \geq 0.$$

SVDErrata

2. Chapter 9: handling with roll motion

Chapter 9 is not a completely new model. It is:

Chapter 3 planar model + roll/pitch motion + roll/pitch inertia + damper terms in load transfer.

The book explicitly says that the Chapter 3 analysis is extended by taking roll and pitch into account, while rigid tyres are assumed as in Sect. 3.8.13.

SVDErrata

2.1 Why Chapter 9 introduces the vehicle invariant point $\bar{M}$

In Chapter 3, u, v are naturally tied to the planar motion of G .

In Chapter 9, if the body rolls, point G moves laterally due to roll motion. So using G to define lateral velocity would mix actual vehicle lateral motion with roll-induced motion.

Chapter 9 therefore introduces the vehicle invariant point, VIP, called $\bar{M}$. The book defines $\bar{M}$ as the point of the vehicle body that, in the reference configuration, coincides with the no-roll center Q . It is chosen as the point best suited to represent vehicle position, lateral velocity, and lateral acceleration when roll is included.

978-94-017-8533-4

So in Chapter 9, read

$$u, \; v$$

as the velocity components of $\bar{M}$, not G :

$$\mathbf{V}_{\bar{M}} = u\hat{\mathbf{i}}_1 + v\hat{\mathbf{j}}_1.$$

Therefore

$$\mathbf{a}_{\bar{M}} = (\dot{u} - vr)\hat{\mathbf{i}}_1 + (\dot{v} + ur)\hat{\mathbf{j}}_1.$$

This is the same-looking formula as Chapter 3, but now applied to $\bar{M}$.

2.2 Yaw-pitch-roll sequence

Chapter 9 defines orientation by the yaw-pitch-roll sequence

$$S_0 \quad \psi \quad S_1 \quad \theta \quad S_2 \quad \phi \quad S_3.$$

The angular velocity components in the body-fixed frame are

$$\begin{aligned}p &= \dot{\phi} - \dot{\psi} \sin \theta, \\q &= \dot{\theta} \cos \phi + \dot{\psi} \sin \phi \cos \theta, \\r &= \dot{\psi} \cos \phi \cos \theta - \dot{\theta} \sin \phi.\end{aligned}$$

For small pitch and roll, the book simplifies these to

$$\begin{aligned}p &\simeq \dot{\phi} - \dot{\psi}\theta, \\q &\simeq \dot{\theta} + \dot{\psi}\phi, \\r &\simeq \dot{\psi}.\end{aligned}$$

The unit-vector approximations are

$$\begin{aligned}\hat{\mathbf{i}}_3 &\simeq \hat{\mathbf{i}}_1 - \theta \mathbf{k}_1, \\\hat{\mathbf{j}}_3 &\simeq \hat{\mathbf{j}}_1 + \phi \mathbf{k}_1, \\\mathbf{k}_3 &\simeq \theta \hat{\mathbf{i}}_1 - \phi \hat{\mathbf{j}}_1 + \mathbf{k}_1.\end{aligned}$$

These are Eq. (9.9)–(9.14).

978-94-017-8533-4

2.3 Position of G relative to $\bar{M}$

Define

$$H = h - d.$$

The book writes this as $h - d$. Here d is the height associated with the VIP/no-roll-center construction in Chapter 9.

Because G is above $\bar{M}$,

$$\mathbf{MG} = H\mathbf{k}_3.$$

Using the small-angle expression for $\mathbf{k}_3$,

$$\mathbf{MG} \simeq H(\theta\hat{\mathbf{i}}_1 - \phi\hat{\mathbf{j}}_1 + \mathbf{k}_1).$$

This is the geometric core of Chapter 9.

It means:

$$\begin{aligned}\text{pitch produces a longitudinal displacement } H\theta, \\ \text{roll produces a lateral displacement } -H\phi.\end{aligned}$$

2.4 Acceleration of G

Use rigid-body kinematics:

$$\mathbf{a}_G = \mathbf{a}_{\bar{M}} + \dot{\boldsymbol{\Omega}} \times \mathbf{MG} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{MG}).$$

Chapter 9 then applies its small-angle/small-rate simplifications and obtains

$$\mathbf{a}_G \simeq \left[(\dot{u} - vr) + H\ddot{\theta} \right] \hat{\mathbf{i}}_1 + \left[(\dot{v} + ur) - H\ddot{\phi} \right] \hat{\mathbf{j}}_1.$$

Therefore define

$$\begin{aligned}a_x &= (\dot{u} - vr) + H\ddot{\theta}, \\a_y &= (\dot{v} + ur) - H\ddot{\phi}.\end{aligned}$$

Compare this to Chapter 3:

$$\begin{aligned}a_x^{\text{Ch.3}} &= \dot{u} - vr, \\a_y^{\text{Ch.3}} &= \dot{v} + ur.\end{aligned}$$

So Chapter 9 adds:

$$+H\ddot{\theta}$$

to longitudinal acceleration and

$$-H\ddot{\phi}$$

to lateral acceleration.

That is the first major Chapter 9 change.

2.5 Chapter 9 translational EOM

Apply

$$m\mathbf{a}_G = \mathbf{F}.$$

Then

$$\begin{aligned}m \left[(\dot{u} - vr) + H\ddot{\theta} \right] &= X, \\m \left[(\dot{v} + ur) - H\ddot{\phi} \right] &= Y, \\0 &= Z.\end{aligned}$$

These are the first three equations of Eq. (9.39).

978-94-017-8533-4

2.6 Angular momentum derivative: where the errata matters

Chapter 9 next derives

$$\dot{\mathbf{K}}_G^r.$$

The printed Eq. (9.31) and Eq. (9.32) contain sign errors. The errata corrects Eq. (9.31) by changing the sign inside the last parentheses of each row, and corrects Eq. (9.32) by changing the sign inside the last parentheses in the first row.

SVDErrata

After applying the errata, and then using the same simplifying assumptions used by the book, the final simplified result is

$$\dot{\mathbf{K}}_G^r = (J_x\ddot{\phi} - J_{xx}\ddot{r})\hat{\mathbf{i}}_1 + J_y\ddot{\theta}\hat{\mathbf{j}}_1 + (J_z\ddot{\theta} - J_{xz}\ddot{\phi})\mathbf{k}_1.$$

This is Eq. (9.36) in the simplified form, but the upstream Eq. (9.31)/(9.32) must be read with the errata.

2.7 Torque equation about $\bar{M}$

The angular momentum equation about $\bar{M}$ is not simply

$$\dot{\mathbf{K}}_G^r = \mathbf{M}_{\bar{M}}.$$

Because $\bar{M}$ is not G , the book uses

$$\dot{\mathbf{K}}_G^r + \mathbf{MG} \times m\mathbf{a}_G = \mathbf{M}_{\bar{M}}.$$

The extra term is

$$\mathbf{MG} \times m\mathbf{a}_G.$$

Using the simplified expressions, the book obtains

$$\mathbf{M}\mathbf{G} \times m\mathbf{a}_G \simeq m \left[H^2\ddot{\phi} - H(\dot{v} + ur) \right] \mathbf{j}_1 + m \left[H^2\ddot{\theta} + H(\dot{u} - vr) \right] \mathbf{j}_1 + mH\dot{u}\phi \mathbf{k}_1,$$

This is Eq. (9.37)/(9.38). 978-94-017-8533-4

2.8 Chapter 9 rotational EOM

Now add

$$\mathbf{\ddot{K}}_G^r$$

and

$$\mathbf{M}\mathbf{G} \times m\mathbf{a}_G.$$

Roll direction:

$$(J_x\ddot{\phi} - J_{xx}\dot{r}) + m \left[H^2\ddot{\phi} - H(\dot{v} + ur) \right] = L_M.$$

Therefore

$$(J_x + mH^2)\ddot{\phi} - J_{xx}\dot{r} - mH(\dot{v} + ur) = L_M.$$

Pitch direction:

$$J_y\ddot{\theta} + m \left[H^2\ddot{\theta} + H(\dot{u} - vr) \right] = M_M.$$

Therefore

$$(J_y + mH^2)\ddot{\theta} + mH(\dot{u} - vr) = M_M.$$

Yaw direction:

$$J_z\dot{r} - J_{xx}\ddot{\phi} + mH\dot{u}\phi = N.$$

So the full Chapter 9 inertial EOM are

$$m \left[(\dot{u} - vr) + H\ddot{\theta} \right] = X,$$

$$m \left[(\dot{v} + ur) - H\ddot{\phi} \right] = Y,$$

$$0 = Z,$$

$$(J_x + mH^2)\ddot{\phi} - J_{xx}\dot{r} - mH(\dot{v} + ur) = L_M,$$

$$(J_y + mH^2)\ddot{\theta} + mH(\dot{u} - vr) = M_M,$$

$$J_z\dot{r} - J_{xx}\ddot{\phi} + mH\dot{u}\phi = N.$$

This is Eq. (9.39), with $H = h - d$. 978-94-017-8533-4

2.9 Chapter 9 right-hand sides

Chapter 9 then completes the model by inserting force and moment resultants.

The stiffness and damping definitions are

$$k_\phi = k_{\phi 1} + k_{\phi 2},$$

$$c_\phi = c_{\phi 1} + c_{\phi 2},$$

and, from the Chapter 8 pitch model,

$$k_\theta = k_1a_1^2 + k_2a_2^2,$$

$$c_\theta = c_1a_1^2 + c_2a_2^2.$$

The Chapter 9 right-hand sides are

$$X = X_1 + X_2 - \frac{1}{2}\rho SC_xu^2,$$

$$Y = Y_1 + Y_2,$$

$$Z = Z_1 + Z_1^q + Z_2 + Z_2^q - mg,$$

$$L_M = -k_\phi\phi - c_\phi\dot{\phi} + mgH\phi,$$

$$M_M = -k_\theta\theta - c_\theta\dot{\theta},$$

$$N = (Y_1a_1 - Y_2a_2) + (\Delta X_1t_1 + \Delta X_2t_2).$$

Again, the vertical-force equation is corrected by the errata: printed Eq. (9.44) repeats Z_1^q , but the corrected equation is

$$Z = Z_1 + Z_1^q + Z_2 + Z_2^q - mg.$$

SVDErrata

2.10 Fully substituted corrected Chapter 9 EOM

Substitute the right-hand sides into Eq. (9.39):

$$m \left[(\dot{u} - vr) + H\ddot{\theta} \right] = X_1 + X_2 - \frac{1}{2}\rho SC_xu^2,$$

$$m \left[(\dot{v} + ur) - H\ddot{\phi} \right] = Y_1 + Y_2,$$

$$Z_1 + Z_1^q + Z_2 + Z_2^q - mg = 0,$$

$$(J_x + mH^2)\ddot{\phi} - J_{xx}\dot{r} - mH(\dot{v} + ur) = -k_\phi\phi - c_\phi\dot{\phi} + mgH\phi,$$

$$(J_y + mH^2)\ddot{\theta} + mH(\dot{u} - vr) = -k_\theta\theta - c_\theta\dot{\theta},$$

$$J_z\dot{r} - J_{xx}\ddot{\phi} + mH\dot{u}\phi = (Y_1a_1 - Y_2a_2) + (\Delta X_1t_1 + \Delta X_2t_2).$$

This is the corrected Chapter 9 EOM set you should use when reading.

3. Chapter 9 tyre vertical loads

Chapter 9 says the tyre constitutive equations are still the Chapter 3 tyre equations; roll and pitch do not directly change the tyre model. What changes is the load-transfer calculation, because now the load transfers depend explicitly on angular accelerations. 978-94-017-8533-4

3.1 Chapter 9 lateral load transfers

Chapter 3 rigid-tyre lateral load transfer was

$$\Delta Z_it_i = Y_iq_i + k_{\phi i}\phi.$$

Chapter 9 adds damper contribution:

$$\Delta Z_1t_1 = Y_1q_1 + k_{\phi 1}\phi + c_{\phi 1}\dot{\phi},$$

$$\Delta Z_2t_2 = Y_2q_2 + k_{\phi 2}\phi + c_{\phi 2}\dot{\phi}.$$

So

$$\Delta Z_1 = \frac{Y_1q_1 + k_{\phi 1}\phi + c_{\phi 1}\dot{\phi}}{t_1},$$

$$\Delta Z_2 = \frac{Y_2q_2 + k_{\phi 2}\phi + c_{\phi 2}\dot{\phi}}{t_2}.$$

This is Eq. (9.45). 978-94-017-8533-4

3.2 Chapter 9 total lateral load-transfer check

Adding the two equations gives

$$\Delta Z_1t_1 + \Delta Z_2t_2 = k_\phi\phi + c_\phi\dot{\phi} + Y_1q_1 + Y_2q_2.$$

The book uses

$$Yd = Y_1q_1 + Y_2q_2.$$

So

$$\Delta Z_1t_1 + \Delta Z_2t_2 = k_\phi\phi + c_\phi\dot{\phi} + Yd.$$

Since $Y = ma_y$,

$$\Delta Z_1t_1 + \Delta Z_2t_2 = k_\phi\phi + c_\phi\dot{\phi} + ma_yd.$$

Combining this with the roll equation gives

$$YH + mgH\phi = k_\phi\phi + c_\phi\dot{\phi} + J_x\ddot{\phi} - J_{xx}\dot{r}.$$

Then the total lateral load transfer becomes

$$\Delta Z_1t_1 + \Delta Z_2t_2 = ma_yh + mgH\phi - (J_x\ddot{\phi} - J_{xx}\dot{r}).$$

This is Eq. (9.46)–(9.48). 978-94-017-8533-4

Do not use this total equation to split front/rear load transfer. The split still comes from

$$\Delta Z_1t_1 = Y_1q_1 + k_{\phi 1}\phi + c_{\phi 1}\dot{\phi},$$

$$\Delta Z_2t_2 = Y_2q_2 + k_{\phi 2}\phi + c_{\phi 2}\dot{\phi}.$$

The total equation is a consistency check.

3.3 Chapter 9 longitudinal load transfer

Chapter 9 generalizes Chapter 3 longitudinal load transfer to include pitch acceleration:

$$\Delta Z = \frac{-Xh + J_\theta\ddot{\theta}}{l}.$$

Since

$$X = ma_x,$$

this is

$$\Delta Z = \frac{-ma_xh + J_\theta\ddot{\theta}}{l}.$$

This is Eq. (9.49). 978-94-017-8533-4

Remember that now

$$a_x = (\dot{u} - vr) + H\ddot{\theta}.$$

3.4 Chapter 9 individual tyre vertical loads

Use the same Chapter 3 definitions:

$$\begin{aligned}Z_{11} &= \frac{Z_1}{2} - \Delta Z_1, \\Z_{12} &= \frac{Z_1}{2} + \Delta Z_1, \\Z_{21} &= \frac{Z_2}{2} - \Delta Z_2, \\Z_{22} &= \frac{Z_2}{2} + \Delta Z_2.\end{aligned}$$

The axle loads are

$$\begin{aligned}Z_1 &= Z_1^0 + \Delta Z - Z_1^a, \\Z_2 &= Z_2^0 - \Delta Z - Z_2^a,\end{aligned}$$

with

$$\begin{aligned}Z_1^0 &= \frac{mga_2}{l}, \\Z_2^0 &= \frac{mga_1}{l}.\end{aligned}$$

Therefore the Chapter 9 tyre vertical loads are

$$\begin{aligned}Z_{11} &= \frac{1}{2} \left[\frac{mga_2}{l} + \frac{-ma_xh + J_y\ddot{\theta}}{l} - Z_1^a \right] - \frac{Y_1q_1 + k_{\delta 1}\phi + c_{\phi 1}\dot{\phi}}{t_1}, \\Z_{12} &= \frac{1}{2} \left[\frac{mga_2}{l} + \frac{-ma_xh + J_y\ddot{\theta}}{l} - Z_1^a \right] + \frac{Y_1q_1 + k_{\delta 1}\phi + c_{\phi 1}\dot{\phi}}{t_1}, \\Z_{21} &= \frac{1}{2} \left[\frac{mga_1}{l} - \frac{-ma_xh + J_y\ddot{\theta}}{l} - Z_2^a \right] - \frac{Y_2q_2 + k_{\delta 2}\phi + c_{\phi 2}\dot{\phi}}{t_2}, \\Z_{22} &= \frac{1}{2} \left[\frac{mga_1}{l} - \frac{-ma_xh + J_y\ddot{\theta}}{l} - Z_2^a \right] + \frac{Y_2q_2 + k_{\delta 2}\phi + c_{\phi 2}\dot{\phi}}{t_2},\end{aligned}$$

where

$$\begin{aligned}a_x &= (\dot{u} - vr) + H\ddot{\theta}, \\H &= h - d.\end{aligned}$$

4. How to read the chapters without getting lost

Use this sequence.

Step 1: Read Chapter 3 Sections 3.1–3.2 only for kinematics

Your goal is only to understand

$$a_x = \dot{u} - vr, \qquad a_y = \dot{v} + ur.$$

Do not worry about tyres yet.

Also apply the errata for the early kinematic parts: β versus $\hat{\beta}$, Eq. (3.5), Eq. (3.15), Eq. (3.32), and Eq. (3.40). These corrections matter for the kinematics discussion even if they are not all directly used in the final EOM.

SVDerrata

Step 2: Read Chapter 3 Sections 3.4–3.6 for equilibrium

Your goal is to understand that the six rigid-body equilibrium equations are split into:

three in-plane EOM

and

three out-of-plane constraint equations.

The in-plane equations become the actual motion equations. The out-of-plane equations determine vertical tyre loads and load transfers. 978-94-017-8531-4

Step 3: Read Chapter 3 Section 3.7 for longitudinal load transfer

Your target is

$$\Delta Z = \frac{-ma_xh + J_{zz}r^2}{l}.$$

Then understand

$$\begin{aligned}Z_1 &= Z_1^0 + \Delta Z - Z_1^a, \\Z_2 &= Z_2^0 - \Delta Z - Z_2^a.\end{aligned}$$

Step 4: Read Chapter 3 Section 3.8 for lateral load transfer

Do not try to memorize the full Eq. (3.113). Instead, understand this internal suspension logic:

$$\begin{aligned}\Delta Z_i t_i &= k_{\delta i}^p \phi_i^p, \\\Delta Z_i t_i &= Y_i q_i + k_{\phi i}^s \phi_i^s, \\\phi &= \phi_i^t + \phi_i^p.\end{aligned}$$

Then the rigid-tyre simplification is

$$\Delta Z_i t_i = Y_i q_i + k_{\delta i} \phi.$$

This is the form Chapter 9 will use.

Step 5: Read Chapter 3 Section 3.11 as the collected model

At this point, the final Chapter 3 model is:

$$\begin{aligned}m(\dot{u} - vr) &= X_1 + X_2 - \frac{1}{2} \rho S C_x u^2, \\m(\dot{v} + ur) &= Y_1 + Y_2, \\J_z \dot{r} &= Y_1 a_1 - Y_2 a_2 + \Delta X_1 t_1 + \Delta X_2 t_2,\end{aligned}$$

plus the tyre vertical loads and tyre constitutive equations.

Also apply the errata to Eq. (3.45), Eq. (3.46), Eq. (3.110), Eq. (3.122), and Eq. (3.126) if you read beyond the EOM.

SVDerrata

Step 6: Read Chapter 9 Sections 9.1–9.5 for the VIP

Your goal is to understand why Chapter 9 changes the point used to define u , v . In Chapter 9:

$$u, v$$

belong to M , not G .

Step 7: Read Chapter 9 Sections 9.2–9.4 for yaw-pitch-roll kinematics

Your target equations are

$$\begin{aligned}p &\simeq \dot{\phi} - \dot{\psi} \theta, \\q &\simeq \dot{\theta} + \dot{\psi} \phi, \\r &\simeq \dot{\psi},\end{aligned}$$

and

$$\mathbf{k}_3 \simeq \theta \mathbf{\hat{1}}_1 - \phi \mathbf{\hat{1}}_1 + \mathbf{k}_1.$$

Step 8: Read Chapter 9 Section 9.6 for the new EOM

Your target is:

$$\mathbf{a}_G \simeq [(\dot{u} - vr) + H\ddot{\theta}] \mathbf{\hat{1}}_1 + [(\dot{v} + ur) - H\dot{\phi}] \mathbf{\hat{1}}_1.$$

Then the force equations follow immediately.

The rotational equations come from

$$\dot{\mathbf{K}}_G + \mathbf{M}\mathbf{G} \times m\mathbf{a}_G = \mathbf{M}\mathbf{\dot{M}}.$$

Apply the errata to Eq. (9.31)/(9.32) before trusting the angular-momentum algebra. SVDerrata

Step 9: Read Chapter 9 Section 9.7 for final force/moment RHS and load transfers

Your target final updates are:

$$\begin{aligned}\Delta Z_i t_i &= Y_i q_i + k_{\phi i} \phi + c_{\phi i} \dot{\phi}, \\\Delta Z &= \frac{-ma_xh + J_y \ddot{\theta}}{l}.\end{aligned}$$

These are the two main tyre-load changes from Chapter 3 to Chapter 9.

5. Common traps to avoid

Trap 1: Confusing G and M .

Chapter 3 uses G . Chapter 9 defines u , v at M , then computes a_G .

Trap 2: Thinking the car physically rolls about a roll axis.

Chapter 9 explicitly argues against that interpretation. M is a useful point for defining motion, not a literal hinge.

Trap 3: Using printed Eq. (9.44) without errata.

Use

$$Z = Z_1 + Z_1^a + Z_2 + Z_2^a - mg.$$

Do not use the repeated- Z_1^a version.

SVDerrata

Trap 4: Using Chapter 3 Eq. (3.110) without errata.

The corrected signs matter if you compute suspension roll-angle components.

SVDerrata

Trap 5: Thinking $\Delta Z_1, \Delta Z_2$ come only from global equilibrium.

Global equilibrium gives the total lateral load-transfer moment. Suspension geometry and stiffness split it between front and rear.

Trap 6: Forgetting the tyre-load sum check.

For Chapter 9,

$$Z_{11} + Z_{12} + Z_{21} + Z_{22} = mg - Z_1^a - Z_2^a.$$

This must match

$$Z_1 + Z_1^a + Z_2 + Z_2^a - mg = 0.$$

If your implementation violates this, the vertical-load equations are wrong.

6. The shortest final synthesis

Chapter 3:

$$m(\dot{u} - vr) = X, \quad m(\dot{v} + ur) = Y, \quad J_z \dot{r} = N$$

with

$$Z_{i1} = \frac{Z_i}{2} - \Delta Z_i, \quad Z_{i2} = \frac{Z_i}{2} + \Delta Z_i$$

and, for rigid tyres,

$$\Delta Z_i t_i = Y_i q_i + k_{\phi i} \phi.$$

Chapter 9 changes the acceleration of G :

$$a_x = (\dot{u} - vr) + (h - d)\ddot{\theta}$$

$$a_y = (\dot{v} + ur) - (h - d)\ddot{\phi}$$

and adds roll/pitch/yaw rotational dynamics:

$$(J_x + m(h - d)^2)\ddot{\phi} - J_{xx}\dot{r} - m(h - d)(\dot{v} + ur) = L_M$$

$$(J_y + m(h - d)^2)\ddot{\theta} + m(h - d)(\dot{u} - vr) = M_M$$

$$J_z \dot{r} - J_{xx}\dot{\phi} + m(h - d)\dot{u}\phi = N.$$

Then Chapter 9 updates tyre loads through

$$\Delta Z_i t_i = Y_i q_i + k_{\phi i} \phi + c_{\phi i} \dot{\phi}$$

and

$$\Delta Z = \frac{-ma_x h + J_y \ddot{\theta}}{l}.$$

That is the clean path through Chapters 3 and 9.

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From the EOMs of Chapter 9 and tyre load equations, if assuming no roll and no pitch and assuming only 7DOF of the vehicle model of a planar car, can you derive the EOMs and tyre load back?

已思考 2m 28s >

Yes. If “no roll and no pitch” means no roll/pitch DOFs in the state vector, then the corrected Chapter 9 model reduces back to the planar Chapter 3-style chassis EOM. But there is one important trap:

You must **not** keep the Chapter 9 roll and pitch dynamic equations after setting $\phi = \theta = 0$. In a planar 7DOF model, roll and pitch are removed as DOFs, so their equations become algebraic load-transfer constraints, not extra motion equations. Chapter 9 explicitly says it extends Chapter 3 by adding roll/pitch and that the tyre equations themselves still come from Chapter 3; the errata also corrects the Chapter 9 vertical force equation $Z = Z_1 + Z_1^q + Z_2 + Z_2^q - mg$. 978-94-017-8533-4 SVDErrata

Let

$$H = h - d.$$

Chapter 9 has

$$a_x = (\dot{u} - vr) + H\ddot{\theta}, \quad a_y = (\dot{v} + ur) - H\ddot{\phi}.$$

For a planar car,

$$\theta = \phi = \dot{\theta} = \dot{\phi} = \ddot{\theta} = \ddot{\phi} = 0,$$

so

$$a_x = \dot{u} - vr, \quad a_y = \dot{v} + ur.$$

1. Planar chassis EOM recovered from Chapter 9

The corrected Chapter 9 force equations become

$$m(\dot{u} - vr) = X,$$

$$m(\dot{v} + ur) = Y,$$

$$0 = Z.$$

Using the Chapter 9/Chapter 3 force resultants,

$$X = X_1 + X_2 - X_a, \quad X_a = \frac{1}{2}\rho SC_x u^2,$$

$$Y = Y_1 + Y_2,$$

$$Z = Z_1 + Z_1^q + Z_2 + Z_2^q - mg.$$

Therefore the planar longitudinal and lateral EOM are

$$m(\dot{u} - vr) = X_1 + X_2 - \frac{1}{2}\rho SC_x u^2$$

$$m(\dot{v} + ur) = Y_1 + Y_2$$

The Chapter 9 yaw equation is

$$J_z \dot{r} - J_{xx}\dot{\phi} + mH\dot{u}\phi = N.$$

With no roll,

$$\phi = 0, \quad \ddot{\phi} = 0,$$

so

$$J_z \dot{r} = N$$

and

$$N = Y_1 a_1 - Y_2 a_2 + \Delta X_1 t_1 + \Delta X_2 t_2.$$

So the recovered planar chassis EOM are

$$m(\dot{u} - vr) = X_1 + X_2 - \frac{1}{2}\rho SC_x u^2$$

$$m(\dot{v} + ur) = Y_1 + Y_2$$

$$J_z \dot{r} = Y_1 a_1 - Y_2 a_2 + \Delta X_1 t_1 + \Delta X_2 t_2$$

This is exactly the Chapter 3 planar handling structure. Chapter 9 even says it is useful to compare its equations with Chapter 3 Eq. (3.64)/(3.65), where pitch/roll inertial effects are neglected. 978-94-017-8533-4

Written as explicit state equations:

$$\dot{u} = vr + \frac{X_1 + X_2 - \frac{1}{2}\rho SC_x u^2}{m}$$

$$\dot{v} = -ur + \frac{Y_1 + Y_2}{m}$$

$$\dot{r} = \frac{Y_1 a_1 - Y_2 a_2 + \Delta X_1 t_1 + \Delta X_2 t_2}{J_z}$$

2. Add the four wheel-spin equations for a 7DOF planar model

The planar chassis has 3 DOF:

$$u, \quad v, \quad r.$$

A usual 7DOF planar vehicle model adds four wheel rotational DOFs:

$$\omega_{11}, \quad \omega_{12}, \quad \omega_{21}, \quad \omega_{22}.$$

So the dynamic state vector is commonly

$$[u, \, v, \, r, \, \omega_{11}, \, \omega_{12}, \, \omega_{21}, \, \omega_{22}]^T.$$

For each wheel, the spin balance is

$$J_{wij}\dot{\omega}_{ij} = T_{ij} - R_i F_{xij}$$

where T_{ij} is the net applied wheel torque, positive in the direction increasing ω_{ij} . If you split drive and brake torques,

$$J_{wij}\dot{\omega}_{ij} = T_{d,ij} - T_{b,ij} - R_i F_{xij}$$

with the sign of F_{xij} interpreted in the wheel longitudinal direction.

So the 7DOF model is:

$$\dot{u} = vr + \frac{X_1 + X_2 - \frac{1}{2}\rho SC_x u^2}{m}$$

$$\dot{v} = -ur + \frac{Y_1 + Y_2}{m}$$

$$\dot{r} = \frac{Y_1 a_1 - Y_2 a_2 + \Delta X_1 t_1 + \Delta X_2 t_2}{J_z}$$

$$J_{w11}\dot{\omega}_{11} = T_{11} - R_1 F_{x11}$$

$$J_{w12}\dot{\omega}_{12} = T_{12} - R_1 F_{x12}$$

$$J_{w21}\dot{\omega}_{21} = T_{21} - R_2 F_{x21}$$

$$J_{w22}\dot{\omega}_{22} = T_{22} - R_2 F_{x22}$$

The position kinematics, if needed, are

$$\dot{\psi} = r,$$

$$\dot{X}_0 = u \cos \psi - v \sin \psi,$$

$$\dot{Y}_0 = u \sin \psi + v \cos \psi.$$

These position equations are not part of the 7 dynamic DOFs, but they are usually integrated to draw the vehicle trajectory.

3. Force aggregation

For each tyre,

$$\begin{aligned}X_{ij} &= F_{xij} \cos \delta_{ij} - F_{yij} \sin \delta_{ij}, \\Y_{ij} &= F_{xij} \sin \delta_{ij} + F_{yij} \cos \delta_{ij}.\end{aligned}$$

Then

$$\begin{aligned}X_1 &= X_{11} + X_{12}, & X_2 &= X_{21} + X_{22}, \\Y_1 &= Y_{11} + Y_{12}, & Y_2 &= Y_{21} + Y_{22}, \\ \Delta X_1 &= \frac{X_{12} - X_{11}}{2}, & \Delta X_2 &= \frac{X_{22} - X_{21}}{2}.\end{aligned}$$

Then the yaw moment is

$$N = Y_1 a_1 - Y_2 a_2 + \Delta X_1 t_1 + \Delta X_2 t_2.$$

4. Longitudinal tyre-load transfer recovered from Chapter 9

Chapter 9 gives the longitudinal load transfer

$$\Delta Z = \frac{-Xh + J_y \ddot{\theta}}{l}.$$

For no pitch,

$$\ddot{\theta} = 0.$$

Since

$$X = ma_x = m(\dot{u} - vr),$$

we get

$$\Delta Z = \frac{-ma_x h}{l} = \frac{-m(\dot{u} - vr)h}{l}$$

So the front and rear axle loads are

$$\begin{aligned}Z_1 &= Z_1^0 + \Delta Z - Z_1^a \\Z_2 &= Z_2^0 - \Delta Z - Z_2^a\end{aligned}$$

with

$$Z_1^0 = \frac{mga_2}{l}, \quad Z_2^0 = \frac{mga_1}{l}.$$

Therefore,

$$\begin{aligned}Z_1 &= \frac{mga_2}{l} - \frac{ma_x h}{l} - Z_1^a \\Z_2 &= \frac{mga_1}{l} + \frac{ma_x h}{l} - Z_2^a\end{aligned}$$

Check the sign: under braking,

$$a_x < 0,$$

so

$$-\frac{ma_x h}{l} > 0.$$

Thus the front load increases and the rear load decreases, which is correct.

Strictly from Chapter 9, this is the planar reduction. If you go back to the fuller Chapter 3 vertical-load formula, there can also be a small $J_{zz}\ddot{r}^2/l$ term in longitudinal load transfer. The book later notes that this term is almost certainly negligible in race-car handling use.

978-94-017-8533-4

5. Lateral tyre-load transfer: the subtle point

Chapter 9 lateral load transfer is

$$\begin{aligned}\Delta Z_1 t_1 &= Y_1 q_1 + k_{\phi 1} \phi + c_{\phi 1} \dot{\phi}, \\ \Delta Z_2 t_2 &= Y_2 q_2 + k_{\phi 2} \phi + c_{\phi 2} \dot{\phi}.\end{aligned}$$

If you literally set

$$\phi = 0, \quad \dot{\phi} = 0,$$

you get

$$\begin{aligned}\Delta Z_1 t_1 &= Y_1 q_1, \\ \Delta Z_2 t_2 &= Y_2 q_2.\end{aligned}$$

But that is not the usual planar 7DOF vehicle load transfer unless you intentionally want to remove the elastic/suspension part of lateral load transfer. It only keeps the geometric no-roll-center contribution.

For a normal planar 7DOF model, the better interpretation is:

roll is not a dynamic DOF, but lateral load transfer still exists algebraically.

So we suppress ϕ as a state, but we keep the algebraic roll moment distribution. From Chapter 9,

$$\Delta Z_1 t_1 + \Delta Z_2 t_2 = k_\phi \phi + c_\phi \dot{\phi} + Yd.$$

Also, in the no-roll-dynamics planar limit, neglecting roll acceleration and the small $J_{zz}\ddot{r}$ term,

$$k_\phi \phi + c_\phi \dot{\phi} \rightarrow Y(h - d).$$

Therefore, with a front/rear roll-moment distribution

$$\lambda_1 = \frac{k_{\phi 1}}{k_\phi}, \quad \lambda_2 = \frac{k_{\phi 2}}{k_\phi}, \quad \lambda_1 + \lambda_2 = 1,$$

the planar lateral load transfers are

$$\begin{aligned}\Delta Z_1 t_1 &= Y_1 q_1 + \lambda_1 Y(h - d) \\ \Delta Z_2 t_2 &= Y_2 q_2 + \lambda_2 Y(h - d)\end{aligned}$$

or

$$\begin{aligned}\Delta Z_1 &= \frac{Y_1 q_1 + \lambda_1 Y(h - d)}{t_1} \\ \Delta Z_2 &= \frac{Y_2 q_2 + \lambda_2 Y(h - d)}{t_2}\end{aligned}$$

with

$$Y = Y_1 + Y_2 = ma_y.$$

This is the clean planar 7DOF version corresponding to Chapter 3's rigid-tyre load-transfer logic. Chapter 9 says the Chapter 3 tyre analysis still applies and that the main difference with roll/pitch included is the explicit angular-acceleration dependence of load transfers; once roll/pitch accelerations are removed, the load-transfer equations collapse back to algebraic planar form.

978-94-017-8533-4

6. Individual tyre vertical loads for the 7DOF planar model

Use the Chapter 3 definitions:

$$\begin{aligned}Z_{11} &= \frac{Z_1}{2} - \Delta Z_1, & Z_{12} &= \frac{Z_1}{2} + \Delta Z_1, \\ Z_{21} &= \frac{Z_2}{2} - \Delta Z_2, & Z_{22} &= \frac{Z_2}{2} + \Delta Z_2.\end{aligned}$$

Substitute the planar axle loads and lateral load transfers:

$$\begin{aligned}Z_{11} &= \frac{1}{2} \left[\frac{mga_2}{l} - \frac{ma_x h}{l} - Z_1^a \right] - \frac{Y_1 q_1 + \lambda_1 Y(h - d)}{t_1} \\ Z_{12} &= \frac{1}{2} \left[\frac{mga_2}{l} - \frac{ma_x h}{l} - Z_1^a \right] + \frac{Y_1 q_1 + \lambda_1 Y(h - d)}{t_1} \\ Z_{21} &= \frac{1}{2} \left[\frac{mga_1}{l} + \frac{ma_x h}{l} - Z_2^a \right] - \frac{Y_2 q_2 + \lambda_2 Y(h - d)}{t_2} \\ Z_{22} &= \frac{1}{2} \left[\frac{mga_1}{l} + \frac{ma_x h}{l} - Z_2^a \right] + \frac{Y_2 q_2 + \lambda_2 Y(h - d)}{t_2}\end{aligned}$$

where

$$a_x = \dot{u} - vr, \quad Y = Y_1 + Y_2 = m(\dot{v} + ur).$$

These are the tyre vertical loads I would use for the planar 7DOF model derived from the corrected Chapter 9/Chapter 3 logic.

7. Sanity checks

Add all four vertical loads:

$$Z_{11} + Z_{12} + Z_{21} + Z_{22} = Z_1 + Z_2.$$

Using the axle-load equations,

$$Z_1 + Z_2 = \frac{mga_2}{l} + \frac{mga_1}{l} - Z_1^a - Z_2^a.$$

Since $l = a_1 + a_2$,

$$Z_{11} + Z_{12} + Z_{21} + Z_{22} = mg - Z_1^a - Z_2^a$$

which matches the corrected vertical equilibrium equation

$$Z_1 + Z_1^a + Z_2 + Z_2^a - mg = 0.$$

Also, the total lateral load-transfer moment is

$$\Delta Z_1 t_1 + \Delta Z_2 t_2 = Y_1 q_1 + Y_2 q_2 + Y(h - d).$$

Since Chapter 9 uses

$$Yd = Y_1 q_1 + Y_2 q_2,$$

we get

$$\Delta Z_1 t_1 + \Delta Z_2 t_2 = Yd + Y(h - d) = Yh.$$

So

$$\Delta Z_1 t_1 + \Delta Z_2 t_2 = Yh = ma_y h$$

which is exactly what a planar no-roll-DOF model should give.

8. Final answer in compact form

The 7DOF planar model recovered from Chapter 9 is:

$$\begin{aligned}\dot{u} &= vr + \frac{X_1 + X_2 - \frac{1}{2}\rho SC_x u^2}{m} \\ \dot{v} &= -ur + \frac{Y_1 + Y_2}{m} \\ \dot{r} &= \frac{Y_1 a_1 - Y_2 a_2 + \Delta X_1 t_1 + \Delta X_2 t_2}{J_z} \\ J_{wij}\dot{\omega}_{ij} &= T_{ij} - R_i F_{xij}, \quad ij \in \{11, 12, 21, 22\}\end{aligned}$$

with tyre vertical loads

$$\begin{aligned}Z_{11} &= \frac{1}{2} \left[\frac{mga_2}{l} - \frac{ma_z h}{l} - Z_1^a \right] - \frac{Y_1 q_1 + \lambda_1 Y(h-d)}{t_1} \\ Z_{12} &= \frac{1}{2} \left[\frac{mga_2}{l} - \frac{ma_z h}{l} - Z_1^a \right] + \frac{Y_1 q_1 + \lambda_1 Y(h-d)}{t_1} \\ Z_{21} &= \frac{1}{2} \left[\frac{mga_1}{l} + \frac{ma_z h}{l} - Z_2^a \right] - \frac{Y_2 q_2 + \lambda_2 Y(h-d)}{t_2} \\ Z_{22} &= \frac{1}{2} \left[\frac{mga_1}{l} + \frac{ma_z h}{l} - Z_2^a \right] + \frac{Y_2 q_2 + \lambda_2 Y(h-d)}{t_2}\end{aligned}$$

where

$$a_x = \dot{u} - vr, \quad Y = Y_1 + Y_2, \quad \lambda_1 + \lambda_2 = 1.$$

That is the correct planar 7DOF reduction. The key is: remove roll/pitch as dynamic DOFs, but do not delete their load-transfer effects.

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Double check your derivations of the 7DOF model and tyre load equations from the corrected EOMs of Chapter 9 and corresponding tyre load equations. Make sure you show your step-by-step derivations in the most detailed way to me, don't jump from Chapter 9 to Chapter 3's equations, but step-by-step.

已思考 2m 34s >

Yes, I double-checked the reduction carefully, and I need to refine one important point from my previous answer:

For the planar 7DOF reduction, you should not literally set $\phi = 0$ inside every Chapter 9 roll equation and then keep the roll equation as a dynamic equation. That would create a false constraint. Instead, the correct interpretation is:

remove roll and pitch as dynamic DOFs, but keep their load-transfer effects algebraically.

This is the same modeling spirit as the book: Chapter 9 extends the Chapter 3 model by adding roll/pitch dynamics, while still assuming rigid tyres in that section; the errata corrects the Chapter 9 vertical resultant to $Z = Z_1 + Z_1^a + Z_2 + Z_2^a - mg$. 978-94-017-8533-4 SVDeivata

Below I derive the planar 7DOF model directly from the corrected Chapter 9 equations, step by step.

1. Start from corrected Chapter 9 notation

Define

$$H = h - d.$$

Here d is the Chapter 9 VIP/no-roll-center height quantity, defined through

$$Yd = Y_1 q_1 + Y_2 q_2.$$

The corrected Chapter 9 acceleration definitions are

$$\begin{aligned}a_x &= (\dot{u} - vr) + H\ddot{\theta}, \\ a_y &= (\dot{v} + ur) - H\ddot{\phi}.\end{aligned}$$

The corrected Chapter 9 translational EOM are

$$\begin{aligned}m \left[(\dot{u} - vr) + H\ddot{\theta} \right] &= X, \\ m \left[(\dot{v} + ur) - H\ddot{\phi} \right] &= Y, \\ 0 &= Z.\end{aligned}$$

The corrected Chapter 9 rotational EOM are

$$\begin{aligned}(J_x + mH^2)\ddot{\phi} - J_{xz}\dot{r} - mH(\dot{v} + ur) &= L_M, \\ (J_y + mH^2)\ddot{\theta} + mH(\dot{u} - vr) &= M_M, \\ J_z\dot{r} - J_{zx}\ddot{\phi} + mH\dot{u}\phi &= N.\end{aligned}$$

Chapter 9 also rewrites the last three equations as

$$\begin{aligned}J_x\ddot{\phi} - J_{xz}\dot{r} - ma_y H &= L_M, \\ J_y\ddot{\theta} + ma_x H &= M_M, \\ J_z\dot{r} - J_{zx}\ddot{\phi} + ma_x H\phi &= N.\end{aligned}$$

This rewritten form is especially useful for the reduction because it separates the chassis accelerations a_x, a_y . 978-94-017-8533-4

2. Apply the planar/no-pitch/no-roll kinematics

For a planar 7DOF car, roll and pitch are not dynamic coordinates. Therefore,

$$\begin{aligned}\theta &= 0, & \dot{\theta} &= 0, & \ddot{\theta} &= 0, \\ \phi &= 0, & \dot{\phi} &= 0, & \ddot{\phi} &= 0.\end{aligned}$$

Therefore the Chapter 9 accelerations reduce to

$$a_x = (\dot{u} - vr) + H \cdot 0,$$

so

$$a_x = \dot{u} - vr$$

and

$$a_y = (\dot{v} + ur) - H \cdot 0,$$

so

$$a_y = \dot{v} + ur.$$

This is the first clean reduction.

3. Reduce the translational EOM

Start from Chapter 9:

$$m \left[(\dot{u} - vr) + H\ddot{\theta} \right] = X.$$

Set

$$\ddot{\theta} = 0.$$

Then

$$m(\dot{u} - vr) = X.$$

Using the Chapter 9 right-hand side

$$X = X_1 + X_2 - \frac{1}{2}\rho SC_x u^2,$$

we get

$$m(\dot{u} - vr) = X_1 + X_2 - \frac{1}{2}\rho SC_x u^2.$$

Now start from

$$m \left[(\dot{v} + ur) - H\ddot{\phi} \right] = Y.$$

Set

$$\ddot{\phi} = 0.$$

Then

$$m(\dot{v} + ur) = Y.$$

Using

$$Y = Y_1 + Y_2,$$

we get

$$m(\dot{v} + ur) = Y_1 + Y_2.$$

So far the planar reduction is exact under the no-roll/no-pitch kinematic assumption.

4. Reduce the yaw EOM

Start from Chapter 9 yaw equation:

$$J_z\dot{r} - J_{zx}\ddot{\phi} + ma_x H\phi = N.$$

Set

$$\phi = 0, \quad \ddot{\phi} = 0.$$

Then

$$J_z \dot{r} = l$$

Using the Chapter 9 yaw moment expression

$$N = (Y_1 a_1 - Y_2 a_2) + (\Delta X_1 t_1 + \Delta X_2 t_2),$$

we get

$$J_z \dot{r} = Y_1 a_1 - Y_2 a_2 + \Delta X_1 t_1 + \Delta X_2 t_2.$$

Therefore the recovered planar chassis EOM are

$$m(\dot{u} - vr) = X_1 + X_2 - \frac{1}{2} \rho S C_x u^2$$

$$m(\dot{v} + ur) = Y_1 + Y_2$$

$$J_z \dot{r} = Y_1 a_1 - Y_2 a_2 + \Delta X_1 t_1 + \Delta X_2 t_2.$$

This is exactly the planar chassis structure.

5. Important warning: do not keep the roll/pitch equations after setting $\phi = \theta = 0$

This is the trap.

If you take the Chapter 9 pitch equation

$$J_\theta \ddot{\theta} + m a_x H = M_M$$

and set

$$\theta = 0, \quad \dot{\theta} = 0, \quad \ddot{\theta} = 0,$$

then

$$m a_x H = 0.$$

That would force

$$a_x = 0,$$

which is obviously not a valid planar car model.

Similarly, if you take the roll equation

$$J_x \ddot{\phi} - J_{xz} \ddot{r} - m a_y H = L_M$$

and set

$$\phi = 0, \quad \dot{\phi} = 0, \quad \ddot{\phi} = 0,$$

then, since $L_M = 0$, you get

$$-J_{xz} \ddot{r} - m a_y H = 0.$$

That would falsely constrain the lateral acceleration.

So the correct reduction is:

when roll and pitch are removed as DOFs, their dynamic equations are not kept as motion equations.

Instead, their physical effects appear through **algebraic load-transfer equations**. This is the key point.

6. Add four wheel-spin equations to make the model 7DOF

The reduced chassis has three dynamic DOFs:

$$u, \quad v, \quad r.$$

A 7DOF planar vehicle model adds four wheel rotational speeds:

$$\omega_{11}, \quad \omega_{12}, \quad \omega_{21}, \quad \omega_{22}.$$

So the dynamic state is

$$\mathbf{x} = [u, \, v, \, r, \, \omega_{11}, \, \omega_{12}, \, \omega_{21}, \, \omega_{22}]^T.$$

The chassis state equations are

$$\dot{u} = vr + \frac{X_1 + X_2 - \frac{1}{2} \rho S C_x u^2}{m}$$

$$\dot{v} = -ur + \frac{Y_1 + Y_2}{m}$$

$$\dot{r} = \frac{Y_1 a_1 - Y_2 a_2 + \Delta X_1 t_1 + \Delta X_2 t_2}{J_z}.$$

For the four wheel spins, use

$$J_{w,ij} \dot{\omega}_{ij} = T_{ij} - R_i F_{xij}, \quad ij \in \{11, 12, 21, 22\}.$$

If drive and brake torques are separated,

$$J_{w,ij} \dot{\omega}_{ij} = T_{d,ij} - T_{b,ij} - R_i F_{xij}.$$

The book's Chapter 3/9 EOM give the chassis force/moment structure; these four wheel-spin equations are the standard wheel rotational balances needed to make the planar chassis model into a 7DOF model.

7. Now derive the tyre vertical loads from Chapter 9

The corrected Chapter 9 vertical resultant is

$$Z = Z_1 + Z_1^a + Z_2 + Z_2^a - mg.$$

The errata explicitly corrects the second aerodynamic term in Eq. (9.44) to Z_2^a . SVOT/mca

Since Chapter 9 has

$$0 = Z,$$

we get

$$Z_1 + Z_1^a + Z_2 + Z_2^a - mg = 0.$$

Therefore

$$Z_1 + Z_2 = mg - Z_1^a - Z_2^a.$$

Now define the static axle loads

$$Z_1^0 = \frac{m g a_2}{l},$$

$$Z_2^0 = \frac{m g a_1}{l},$$

where

$$l = a_1 + a_2.$$

Then introduce the longitudinal load transfer ΔZ by

$$Z_1 = Z_1^0 + \Delta Z - Z_1^a,$$

$$Z_2 = Z_2^0 - \Delta Z - Z_2^a.$$

These automatically satisfy the vertical-force equation because

$$Z_1 + Z_2 = Z_1^0 + Z_2^0 - Z_1^a - Z_2^a = mg - Z_1^a - Z_2^a.$$

8. Derive longitudinal load transfer from Chapter 9

Chapter 9 gives

$$\Delta Z = \frac{-Xh + J_\theta \ddot{\theta}}{l}.$$

For no pitch DOF,

$$\ddot{\theta} = 0.$$

Therefore

$$\Delta Z = \frac{-Xh}{l}.$$

But from the reduced longitudinal EOM,

$$X = m a_x,$$

and

$$a_x = \dot{u} - vr.$$

Therefore

$$\Delta Z = -\frac{m a_x h}{l} = -\frac{m(\dot{u} - vr)h}{l}.$$

So the front and rear axle loads become

$$Z_1 = \frac{m g a_2}{l} - \frac{m a_x h}{l} - Z_1^a,$$

$$Z_2 = \frac{m g a_1}{l} + \frac{m a_x h}{l} - Z_2^a.$$

Therefore

$$Z_1 = \frac{m g a_2}{l} - \frac{m a_x h}{l} - Z_1^a$$

$$Z_2 = \frac{m g a_1}{l} + \frac{m a_x h}{l} - Z_2^a.$$

Check braking:

$$a_x < 0$$

so

$$-m a_x h > 0,$$

therefore Z_1 increases and Z_2 decreases. Correct.

Strictly from Chapter 9, this is the longitudinal load transfer. If you compare with Chapter 3, Chapter 3 also has a small $J_{xx} r^2$ term in the pitch/vertical-load relation, but the book notes those J_{xx} terms are usually

9. Derive lateral load transfer from Chapter 9 carefully

Chapter 9 gives

$$\begin{aligned}\Delta Z_1 t_1 &= Y_1 q_1 + k_{\phi 1} \phi + c_{\phi 1} \dot{\phi}, \\ \Delta Z_2 t_2 &= Y_2 q_2 + k_{\phi 2} \phi + c_{\phi 2} \dot{\phi}.\end{aligned}$$

Chapter 9 also gives

$$k_\phi = k_{\phi 1} + k_{\phi 2}, \quad c_\phi = c_{\phi 1} + c_{\phi 2}.$$

Therefore, adding both axle equations gives

$$\Delta Z_1 t_1 + \Delta Z_2 t_2 = Y_1 q_1 + Y_2 q_2 + k_\phi \phi + c_\phi \dot{\phi}.$$

Since Chapter 9 uses

$$Y d = Y_1 q_1 + Y_2 q_2,$$

we have

$$\Delta Z_1 t_1 + \Delta Z_2 t_2 = Y d + k_\phi \phi + c_\phi \dot{\phi}.$$

Chapter 9 then combines the roll equation with the load-transfer equation and obtains

$$Y(h-d) + mg(h-d)\phi = k_\phi \phi + c_\phi \dot{\phi} + J_x \ddot{\phi} - J_{xx} \dot{r}.$$

Using $H = h - d$, this is

$$YH + mgH\phi = k_\phi \phi + c_\phi \dot{\phi} + J_x \ddot{\phi} - J_{xx} \dot{r}.$$

This is the key equation for the planar reduction. 978-94-017-8533-4

10. What “no roll DOF” really means for load transfer

For a planar 7DOF model,

$$\phi, \dot{\phi}, \ddot{\phi}$$

are not states. But the load transfer due to lateral acceleration must still exist.

So we take the zero-roll-angle / no-roll-dynamics limit:

$$\phi \rightarrow 0, \quad \dot{\phi} \rightarrow 0, \quad \ddot{\phi} \rightarrow 0,$$

but we do **not** force

$$k_\phi \phi + c_\phi \dot{\phi}$$

to be zero. Instead, that quantity becomes the algebraic roll-resisting moment that produces lateral load transfer.

Define

$$M_\phi = k_\phi \phi + c_\phi \dot{\phi}.$$

Then the Chapter 9 roll/load-transfer equation becomes

$$YH + mgH\phi = M_\phi + J_x \ddot{\phi} - J_{xx} \dot{r}.$$

Now take the planar no-roll limit:

$$\phi = 0, \quad \ddot{\phi} = 0.$$

Then

$$YH = M_\phi - J_{xx} \dot{r}.$$

Therefore

$$M_\phi = YH + J_{xx} \dot{r}.$$

If, as the book often says, J_{xx} effects are negligible, then

$$M_\phi \simeq YH = Y(h-d).$$

This is the precise step that was missing if one simply set $\phi = 0$ inside Eq. (9.45). The elastic/constraint roll moment remains finite even though the roll angle is not a state.

11. Split the algebraic roll-resisting moment front/rear

Chapter 9’s axle load-transfer equations are

$$\begin{aligned}\Delta Z_1 t_1 &= Y_1 q_1 + \left(k_{\phi 1} \phi + c_{\phi 1} \dot{\phi}\right), \\ \Delta Z_2 t_2 &= Y_2 q_2 + \left(k_{\phi 2} \phi + c_{\phi 2} \dot{\phi}\right).\end{aligned}$$

Define

$$\begin{aligned}M_{\phi 1} &= k_{\phi 1} \phi + c_{\phi 1} \dot{\phi}, \\ M_{\phi 2} &= k_{\phi 2} \phi + c_{\phi 2} \dot{\phi}.\end{aligned}$$

Then

$$M_\phi = M_{\phi 1} + M_{\phi 2}.$$

For the planar algebraic reduction, choose the same front/rear roll-moment distribution implied by the book’s roll stiffness split:

$$\lambda_1 = \frac{k_{\phi 1}}{k_\phi}, \quad \lambda_2 = \frac{k_{\phi 2}}{k_\phi}, \quad \lambda_1 + \lambda_2 = 1.$$

Then

$$\begin{aligned}M_{\phi 1} &= \lambda_1 M_\phi, \\ M_{\phi 2} &= \lambda_2 M_\phi.\end{aligned}$$

Using

$$M_\phi = YH + J_{xx} \dot{r},$$

we get

$$\begin{aligned}M_{\phi 1} &= \lambda_1 (YH + J_{xx} \dot{r}), \\ M_{\phi 2} &= \lambda_2 (YH + J_{xx} \dot{r}).\end{aligned}$$

Therefore the planar lateral load transfers derived from Chapter 9 are

$$\begin{aligned}\Delta Z_1 t_1 &= Y_1 q_1 + \lambda_1 (YH + J_{xx} \dot{r}) \\ \Delta Z_2 t_2 &= Y_2 q_2 + \lambda_2 (YH + J_{xx} \dot{r}).\end{aligned}$$

Equivalently,

$$\begin{aligned}\Delta Z_1 &= \frac{Y_1 q_1 + \lambda_1 (YH + J_{xx} \dot{r})}{t_1} \\ \Delta Z_2 &= \frac{Y_2 q_2 + \lambda_2 (YH + J_{xx} \dot{r})}{t_2}.\end{aligned}$$

If $J_{xx} \dot{r}$ is neglected,

$$\begin{aligned}\Delta Z_1 &\simeq \frac{Y_1 q_1 + \lambda_1 Y(h-d)}{t_1} \\ \Delta Z_2 &\simeq \frac{Y_2 q_2 + \lambda_2 Y(h-d)}{t_2}.\end{aligned}$$

This is the clean planar 7DOF lateral load-transfer result.

12. Sanity check for lateral load transfer

Add the two axle load-transfer moments:

$$\Delta Z_1 t_1 + \Delta Z_2 t_2 = Y_1 q_1 + Y_2 q_2 + (\lambda_1 + \lambda_2)(YH + J_{xx} \dot{r}).$$

Since

$$\lambda_1 + \lambda_2 = 1,$$

and

$$Y_1 q_1 + Y_2 q_2 = Y d,$$

we get

$$\Delta Z_1 t_1 + \Delta Z_2 t_2 = Y d + YH + J_{xx} \dot{r}.$$

Since

$$H = h - d,$$

then

$$Y d + YH = Y d + Y(h-d) = Yh.$$

Therefore

$$\Delta Z_1 t_1 + \Delta Z_2 t_2 = Yh + J_{xx} \dot{r}.$$

If $J_{xx} \dot{r}$ is negligible,

$$\Delta Z_1 t_1 + \Delta Z_2 t_2 \simeq Yh.$$

This is exactly the expected planar lateral load-transfer moment. It also matches the Chapter 3 global vertical-load relation, where the lateral load-transfer sum is $ma_y h + J_{xx} \dot{r}$. 978-94-017-8533-4

13. Individual tyre vertical loads

Use the definitions

$$\begin{aligned}Z_{11} &= \frac{Z_1}{2} - \Delta Z_1, \\ Z_{12} &= \frac{Z_1}{2} + \Delta Z_1, \\ Z_{21} &= \frac{Z_2}{2} - \Delta Z_2,\end{aligned}$$

$$Z_{22} = \frac{Z_2}{2} + \Delta Z_2,$$

Now substitute

$$Z_1 = \frac{mga_2 - ma_z h}{l} - Z_1^a,$$

$$Z_2 = \frac{mga_1 + ma_z h}{l} - Z_2^a,$$

and

$$\Delta Z_1 = \frac{Y_1 q_1 + \lambda_1(YH + J_{xz}\dot{r})}{t_1},$$

$$\Delta Z_2 = \frac{Y_2 q_2 + \lambda_2(YH + J_{xz}\dot{r})}{t_2}.$$

So the fully derived planar 7DOF tyre vertical loads are

$$Z_{11} = \frac{1}{2} \left[\frac{mga_2 - ma_z h}{l} - Z_1^a \right] - \frac{Y_1 q_1 + \lambda_1(YH + J_{xz}\dot{r})}{t_1}$$

$$Z_{12} = \frac{1}{2} \left[\frac{mga_2 - ma_z h}{l} - Z_1^a \right] + \frac{Y_1 q_1 + \lambda_1(YH + J_{xz}\dot{r})}{t_1}$$

$$Z_{21} = \frac{1}{2} \left[\frac{mga_1 + ma_z h}{l} - Z_2^a \right] - \frac{Y_2 q_2 + \lambda_2(YH + J_{xz}\dot{r})}{t_2}$$

$$Z_{22} = \frac{1}{2} \left[\frac{mga_1 + ma_z h}{l} - Z_2^a \right] + \frac{Y_2 q_2 + \lambda_2(YH + J_{xz}\dot{r})}{t_2}.$$

where

$$H = h - d,$$

$$a_x = \dot{u} - vr,$$

$$Y = Y_1 + Y_2 = m(\dot{v} + ur).$$

If $J_{xz}\dot{r}$ is neglected, use

$$YH + J_{xz}\dot{r} \approx Y(h - d).$$

Then

$$Z_{11} \simeq \frac{1}{2} \left[\frac{mga_2 - ma_z h}{l} - Z_1^a \right] - \frac{Y_1 q_1 + \lambda_1 Y(h - d)}{t_1}$$

$$Z_{12} \simeq \frac{1}{2} \left[\frac{mga_2 - ma_z h}{l} - Z_1^a \right] + \frac{Y_1 q_1 + \lambda_1 Y(h - d)}{t_1}$$

$$Z_{21} \simeq \frac{1}{2} \left[\frac{mga_1 + ma_z h}{l} - Z_2^a \right] - \frac{Y_2 q_2 + \lambda_2 Y(h - d)}{t_2}$$

$$Z_{22} \simeq \frac{1}{2} \left[\frac{mga_1 + ma_z h}{l} - Z_2^a \right] + \frac{Y_2 q_2 + \lambda_2 Y(h - d)}{t_2}.$$

14. Final 7DOF planar model

The complete dynamic equations are

$$\dot{u} = vr + \frac{X_1 + X_2}{m} - \frac{1}{2}\rho SC_x u^2$$

$$\dot{v} = -ur + \frac{Y_1 + Y_2}{m}$$

$$\dot{r} = \frac{Y_1 a_1 - Y_2 a_2 + \Delta X_1 t_1 + \Delta X_2 t_2}{J_z}$$

$$J_{w,11}\dot{\omega}_{11} = T_{11} - R_1 F_{x11}$$

$$J_{w,12}\dot{\omega}_{12} = T_{12} - R_1 F_{x12}$$

$$J_{w,21}\dot{\omega}_{21} = T_{21} - R_2 F_{x21}$$

$$J_{w,22}\dot{\omega}_{22} = T_{22} - R_2 F_{x22}.$$

The force aggregation is

$$X_i = X_{i1} + X_{i2}, \qquad Y_i = Y_{i1} + Y_{i2},$$

$$\Delta X_i = \frac{X_{i2} - X_{i1}}{2}.$$

The tyre-frame to vehicle-frame transformation is

$$X_{ij} = F_{xij} \cos \delta_{ij} - F_{yij} \sin \delta_{ij},$$

$$Y_{ij} = F_{xij} \sin \delta_{ij} + F_{yij} \cos \delta_{ij}.$$

15. Final double-check conclusions

The corrected reduction is:

Chassis EOM: correct.

$$\text{Longitudinal load transfer: } \Delta Z = -\frac{ma_z h}{l}.$$

$$\text{Lateral load transfer: } \Delta Z_i t_i = Y_i q_i + \lambda_i(Y(h - d) + J_{xz}\dot{r}).$$

Neglecting $J_{xz}\dot{r}$,

$$\Delta Z_i t_i \simeq Y_i q_i + \lambda_i Y(h - d).$$

The most important correction to my previous answer is that the more complete planar reduction from Chapter 9 keeps the possible $J_{xz}\dot{r}$ contribution in lateral load transfer:

$$Y(h - d) + J_{xz}\dot{r}.$$

It is often negligible, but if you want the most faithful reduction from the corrected Chapter 9 load-transfer relation, keep it until you intentionally drop it.


[資料來源](#)

Now you have the corrected EOMs and tyre load equations of Chapter 9 and also the EOMs and tyre load equations of the 7DOF model, tell me the meanings of each variables in them. Then give me the complete list of equations of these two models so that I can use them in the future.

已思考 46s ›

Yes. Below is a clean reference sheet for the corrected Chapter 9 model and the planar 7DOF model reduced from it. I will use the same notation we have been using:

$$H = h - d$$

where H is the vertical distance from the VIP/no-roll-center reference point $\bar{M}$ to the CG G . In the book this is written as $h - d$. Chapter 9 uses the VIP because, with roll motion, the CG G is not the cleanest point for defining lateral velocity; the book explicitly identifies the VIP as the best point for vehicle position, lateral velocity, and lateral acceleration, and assumes rigid tyres in Chapter 9. 978-94-017-8533-4978-94-017-8533-4

The errata must be applied: Chapter 9 Eq. (9.31)/(9.32) have angular-momentum sign corrections, and Eq. (9.44) must use Z_1^a and Z_2^a , not Z_1^a twice. SVDErrata

1. Index convention

Use

$$i = 1 \quad \text{front axle}, \qquad i = 2 \quad \text{rear axle}.$$

Use

$$j = 1, 2$$

for the two sides of each axle. I will call them **side 1** and **side 2**, because the equations only require consistency with the book's sign convention.

Thus:

$$11 = \text{front side 1}, \quad 12 = \text{front side 2},$$

$$21 = \text{rear side 1}, \quad 22 = \text{rear side 2}.$$

The lateral load transfer convention is

$$\Delta Z_i = \frac{Z_{i2} - Z_{i1}}{2}.$$

So positive ΔZ_i means side 2 load increases and side 1 load decreases.

2. Variable meanings

2.1 Time derivatives

An overdot means time derivative:

$$\dot{u} = \frac{du}{dt}, \qquad \ddot{\phi} = \frac{d^2\phi}{dt^2}.$$

2.2 Vehicle motion variables

u

Longitudinal velocity component of the vehicle reference point. In Chapter 3 planar motion it is naturally tied to $\dot{G}$; in Chapter 9 it is defined at the VIP $\bar{M}$.

v

Lateral velocity component of the vehicle reference point. In Chapter 9 it is specifically defined at $\bar{M}$, not G .

r

Yaw-rate component. In the small pitch/roll approximation,

	$r \simeq \dot{\psi}$.
	ψ
Yaw angle.	
	θ
Pitch angle.	
	ϕ
Roll angle.	
	$\dot{\theta}, \ddot{\theta}$
Pitch rate and pitch acceleration.	
	$\dot{\phi}, \ddot{\phi}$
Roll rate and roll acceleration.	
	a_x
Longitudinal acceleration of G , in the vehicle yawed frame.	
For Chapter 9:	$a_x = (\dot{u} - vr) + H\ddot{\theta}$.
For the planar 7DOF reduction:	$a_x = \dot{u} - vr$.
	a_y
Lateral acceleration of G , in the vehicle yawed frame.	
For Chapter 9:	$a_y = (\dot{v} + ur) - H\ddot{\phi}$.
For the planar 7DOF reduction:	$a_y = \dot{v} + ur$.

2.3 Geometry variables

	G
Vehicle center of gravity.	
	M
Vehicle invariant point, VIP. Chapter 9 uses M as the point for defining vehicle position and lateral velocity when roll is included.	
	h
Height of the vehicle CG G above the road plane.	
	d
Height parameter of the VIP/no-roll-center construction used in Chapter 9. It is defined through the relation	$Yd = Y_1q_1 + Y_2q_2$.
	$H = h - d$
Vertical distance from M to G .	
	a_1
Longitudinal distance from G to the front axle.	
	a_2
Longitudinal distance from G to the rear axle.	$l = a_1 + a_2$
Wheelbase.	
	t_1
Front track width.	
	t_2
Rear track width.	
	q_1
Front no-roll-center height parameter used in the lateral load-transfer equations.	
	q_2
Rear no-roll-center height parameter used in the lateral load-transfer equations.	

2.4 Mass and inertia variables

	m
Vehicle mass.	
	g
Gravitational acceleration.	
	J_x
Roll moment of inertia of the vehicle body about G .	
	J_y
Pitch moment of inertia about G .	
	J_z
Yaw moment of inertia about G .	
	J_{xx}
Product of inertia coupling yaw and roll. The errata corrections are important for the angular-momentum terms involving J_{xx} .	SVDerrata
	$J_{w,ij}$
Rotational inertia of wheel i_j .	

2.5 Suspension variables

	$k_{\phi 1}, \quad k_{\phi 2}$
Front and rear roll stiffness contributions.	
	$k_{\phi} = k_{\phi 1} + k_{\phi 2}$
Total roll stiffness.	
	$c_{\phi 1}, \quad c_{\phi 2}$
Front and rear roll damping contributions.	
	$c_{\phi} = c_{\phi 1} + c_{\phi 2}$
Total roll damping.	
	k_{θ}
Pitch stiffness.	
	c_{θ}
Pitch damping.	
In the book's Chapter 9 notation:	$k_{\theta} = k_1a_1^2 + k_2a_2^2$, $c_{\theta} = c_1a_1^2 + c_2a_2^2$, λ_1, λ_2
Front/rear distribution of the algebraic roll-resisting moment in the planar 7DOF reduction:	
	$\lambda_1 = \frac{k_{\phi 1}}{k_{\phi}}, \quad \lambda_2 = \frac{k_{\phi 2}}{k_{\phi}}, \quad \lambda_1 + \lambda_2 = 1$.

2.6 Force and moment variables

	X
Total longitudinal external force on the vehicle.	
	Y
Total lateral external force on the vehicle.	
	Z
Total vertical force residual. In the corrected Chapter 9 equation,	$Z = Z_1 + Z_1^a + Z_2 + Z_2^a - mg$.
The errata corrects the second aerodynamic term to Z_2^a .	SVDerrata
	L_M
Roll moment about the VIP M .	
	M_M
Pitch moment about M .	
	N
Yaw moment.	
	X_i
Total longitudinal road-tire force at axle i , in the vehicle frame.	
	Y_i

Total lateral road-tire force at axle i , in the vehicle frame.

$$X_{ij}, \quad Y_{ij}$$

Vehicle-frame longitudinal and lateral components of the force at tyre ij .

$$F_{xij}, \quad F_{yij}$$

Tyre-frame longitudinal and lateral forces at tyre ij .

$$Z_{ij} = F_{zij}$$

Vertical road normal load at tyre ij .

$$Z_i$$

Total vertical road normal load at axle i :

$$Z_1 = Z_{11} + Z_{12}, \quad Z_2 = Z_{21} + Z_{22}.$$

$$Z_i^0$$

Static axle load:

$$Z_1^0 = \frac{mga_2}{l}, \quad Z_2^0 = \frac{mga_1}{l}.$$

$$Z_i^a$$

Vertical aerodynamic force contribution at axle i , in the book's sign convention. Since the vertical equilibrium is written as

$$Z_1 + Z_1^a + Z_2 + Z_2^a - mg = 0,$$

a negative Z_i^a increases tyre normal load, corresponding to downforce.

$$X_a$$

Aerodynamic drag:

$$X_a = \frac{1}{2} \rho S C_x u^2.$$

$$\rho$$

Air density.

$$S$$

Reference aerodynamic area.

$$C_x$$

Aerodynamic drag coefficient.

2.7 Load-transfer variables

$$\Delta Z$$

Longitudinal load transfer. Positive ΔZ increases front axle load and decreases rear axle load.

$$\Delta Z_1$$

Front lateral load transfer:

$$\Delta Z_1 = \frac{Z_{12} - Z_{11}}{2}.$$

$$\Delta Z_2$$

Rear lateral load transfer:

$$\Delta Z_2 = \frac{Z_{22} - Z_{21}}{2}.$$

$$\Delta X_1, \quad \Delta X_2$$

Left-right longitudinal force differences at front and rear axles, using the same side-1/side-2 convention:

$$\Delta X_1 = \frac{X_{12} - X_{11}}{2}, \quad \Delta X_2 = \frac{X_{22} - X_{21}}{2}.$$

2.8 Steering, tyre, and wheel variables

$$\delta_{ij}$$

Steering angle of wheel ij .

$$\omega_{ij}$$

Angular speed of wheel ij .

$$R_i$$

Effective rolling radius for axle i . Use R_1 for front tyres and R_2 for rear tyres if they differ.

$$T_{ij}$$

Net applied torque at wheel ij . If drive and brake torque are separated,

$$T_{ij} = T_{d,ij} - T_{b,ij}.$$

$$T_{d,ij}$$

Drive torque at wheel ij .

$$T_{b,ij}$$

Brake torque at wheel ij .

$$\sigma_{xij}, \quad \sigma_{yij}$$

Longitudinal and lateral tyre slips.

$$\gamma_{ij}$$

Camber angle of tyre ij .

$$\varphi_{ij}$$

Spin slip or spin-related tyre variable in the book's general tyre-force notation.

3. Corrected Chapter 9 model: complete equation list

This is the corrected Chapter 9 handling-with-roll model, under the book's assumptions: small pitch/roll, flat road, rigid tyres in Chapter 9, and the corrected errata for angular momentum and vertical force. Chapter 9 uses yaw-pitch-roll kinematics; it gives p, q, r as body-frame angular velocity components and simplifies them for small pitch and roll.

978-94-017-8533-4

3.1 Acceleration definitions

$$H = h - d.$$

$$a_x = (\dot{u} - vr) + H\ddot{\theta}.$$

$$a_y = (\dot{v} + ur) - H\ddot{\phi}.$$

3.2 Corrected Chapter 9 EOM

$$m \left[(\dot{u} - vr) + H\ddot{\theta} \right] = X.$$

$$m \left[(\dot{v} + ur) - H\ddot{\phi} \right] = Y.$$

$$0 = Z.$$

$$(J_x + mH^2)\ddot{\phi} - J_{xz}\dot{r} - mH(\dot{v} + ur) = L_M.$$

$$(J_y + mH^2)\ddot{\theta} + mH(\dot{u} - vr) = M_M.$$

$$J_z\dot{r} - J_{xz}\dot{\phi} + mH\dot{u}\phi = N.$$

Equivalent compact rotational form:

$$J_x\ddot{\phi} - J_{xz}\dot{r} - ma_yH = L_M.$$

$$J_y\ddot{\theta} + ma_xH = M_M.$$

$$J_z\dot{r} - J_{xz}\dot{\phi} + ma_xH\phi = N.$$

3.3 Corrected force and moment right-hand sides

$$X = X_1 + X_2 - \frac{1}{2} \rho S C_x u^2.$$

$$Y = Y_1 + Y_2.$$

$$Z = Z_1 + Z_1^a + Z_2 + Z_2^a - mg.$$

The equation above is the corrected version of Chapter 9 Eq. (9.44); the errata says the second aerodynamic vertical term must be Z_2^a .

SVGErrata

$$L_M = -k_\phi\phi - c_\phi\dot{\phi} + mgH\phi.$$

$$M_M = -k_\theta\theta - c_\theta\dot{\theta}.$$

$$N = Y_1a_1 - Y_2a_2 + \Delta X_1t_1 + \Delta X_2t_2.$$

3.4 Fully substituted corrected Chapter 9 EOM

$$m \left[(\dot{u} - vr) + H\ddot{\theta} \right] = X_1 + X_2 - \frac{1}{2} \rho S C_x u^2,$$

$$m \left[(\dot{v} + ur) - H\ddot{\phi} \right] = Y_1 + Y_2.$$

$$Z_1 + Z_1^a + Z_2 + Z_2^a - mg = 0.$$

$$(J_x + mH^2)\ddot{\phi} - J_{xz}\dot{r} - mH(\dot{v} + ur) = -k_\phi\phi - c_\phi\dot{\phi} + mgH\phi.$$

$$(J_y + mH^2)\ddot{\theta} + mH(\dot{u} - vr) = -k_\theta\theta - c_\theta\dot{\theta}.$$

$$J_z\dot{r} - J_{xz}\dot{\phi} + mH\dot{u}\phi = Y_1a_1 - Y_2a_2 + \Delta X_1t_1 + \Delta X_2t_2.$$

3.5 Force aggregation and tyre-frame transformation

For each wheel:

$$X_{ij} = F_{xij} \cos \delta_{ij} - F_{yij} \sin \delta_{ij}.$$

$$Y_{ij} = F_{xij} \sin \delta_{ij} + F_{yij} \cos \delta_{ij}.$$

Axle sums:

$$\begin{aligned} X_1 &= X_{11} + X_{12}, & X_2 &= X_{21} + X_{22}. \\ Y_1 &= Y_{11} + Y_{12}, & Y_2 &= Y_{21} + Y_{22}. \end{aligned}$$

Longitudinal force differences:

$$\begin{aligned} \Delta X_1 &= \frac{X_{12} - X_{11}}{2}, \\ \Delta X_2 &= \frac{X_{22} - X_{21}}{2}. \end{aligned}$$

3.6 Chapter 9 load-transfer equations

Longitudinal load transfer:

$$\Delta Z = -\frac{Xh}{l} + J_y \ddot{\theta}.$$

Since $X = ma_x$,

$$\Delta Z = -\frac{ma_x h}{l} + J_y \ddot{\theta}.$$

Lateral load transfers:

$$\begin{aligned} \Delta Z_1 t_1 &= Y_1 q_1 + k_{\phi 1} \phi + c_{\phi 1} \dot{\phi}, \\ \Delta Z_2 t_2 &= Y_2 q_2 + k_{\phi 2} \phi + c_{\phi 2} \dot{\phi}. \end{aligned}$$

Therefore:

$$\begin{aligned} \Delta Z_1 &= \frac{Y_1 q_1 + k_{\phi 1} \phi + c_{\phi 1} \dot{\phi}}{t_1}, \\ \Delta Z_2 &= \frac{Y_2 q_2 + k_{\phi 2} \phi + c_{\phi 2} \dot{\phi}}{t_2}. \end{aligned}$$

Total lateral load-transfer check:

$$\Delta Z_1 t_1 + \Delta Z_2 t_2 = Yd + k_\phi \phi + c_\phi \dot{\phi}.$$

Using the roll equation:

$$\Delta Z_1 t_1 + \Delta Z_2 t_2 = ma_y h + mgH\phi - (J_\phi \ddot{\phi} - J_{sx} \dot{r}).$$

3.7 Chapter 9 axle loads

Static axle loads:

$$\begin{aligned} Z_1^0 &= \frac{mga_2}{l}, \\ Z_2^0 &= \frac{mga_1}{l}. \end{aligned}$$

Dynamic axle loads:

$$\begin{aligned} Z_1 &= Z_1^0 + \Delta Z - Z_1^a, \\ Z_2 &= Z_2^0 - \Delta Z - Z_2^a. \end{aligned}$$

3.8 Chapter 9 individual tyre vertical loads

General form:

$$\begin{aligned} Z_{11} &= \frac{Z_1}{2} - \Delta Z_1, \\ Z_{12} &= \frac{Z_1}{2} + \Delta Z_1, \\ Z_{21} &= \frac{Z_2}{2} - \Delta Z_2, \\ Z_{22} &= \frac{Z_2}{2} + \Delta Z_2. \end{aligned}$$

Fully expanded:

$$\begin{aligned} Z_{11} &= \frac{1}{2} \left[\frac{mga_2}{l} + \frac{-ma_x h}{l} + J_y \ddot{\theta} - Z_1^a \right] - \frac{Y_1 q_1 + k_{\phi 1} \phi + c_{\phi 1} \dot{\phi}}{t_1}, \\ Z_{12} &= \frac{1}{2} \left[\frac{mga_2}{l} + \frac{-ma_x h}{l} + J_y \ddot{\theta} - Z_1^a \right] + \frac{Y_1 q_1 + k_{\phi 1} \phi + c_{\phi 1} \dot{\phi}}{t_1}, \\ Z_{21} &= \frac{1}{2} \left[\frac{mga_1}{l} - \frac{-ma_x h}{l} + J_y \ddot{\theta} - Z_2^a \right] - \frac{Y_2 q_2 + k_{\phi 2} \phi + c_{\phi 2} \dot{\phi}}{t_2}, \\ Z_{22} &= \frac{1}{2} \left[\frac{mga_1}{l} - \frac{-ma_x h}{l} + J_y \ddot{\theta} - Z_2^a \right] + \frac{Y_2 q_2 + k_{\phi 2} \phi + c_{\phi 2} \dot{\phi}}{t_2}. \end{aligned}$$

Sanity check:

$$Z_{11} + Z_{12} + Z_{21} + Z_{22} = mg - Z_1^a - Z_2^a.$$

4. Planar 7DOF model: complete equation list

The planar 7DOF model is obtained by removing roll and pitch as dynamic DOFs:

$$\begin{aligned} \phi &= 0, & \dot{\phi} &= 0, & \ddot{\phi} &= 0, \\ \theta &= 0, & \dot{\theta} &= 0, & \ddot{\theta} &= 0. \end{aligned}$$

Then

$$\begin{aligned} a_x &= \dot{u} - vr, \\ a_y &= \dot{v} + ur. \end{aligned}$$

The dynamic states are

$$x = [u, \, v, \, r, \, \omega_{11}, \, \omega_{12}, \, \omega_{21}, \, \omega_{22}]^T.$$

Position variables may also be integrated, but they are not part of the 7 dynamic DOFs:

$$\begin{aligned} \dot{\psi} &= r, \\ \dot{X}_0 &= u \cos \psi - v \sin \psi, \\ \dot{Y}_0 &= u \sin \psi + v \cos \psi. \end{aligned}$$

4.1 Planar 7DOF chassis EOM

$$\begin{aligned} m(\dot{u} - vr) &= X_1 + X_2 - \frac{1}{2} \rho S C_x u^2, \\ m(\dot{v} + ur) &= Y_1 + Y_2, \\ J_z \dot{r} &= Y_1 a_1 - Y_2 a_2 + \Delta X_1 t_1 + \Delta X_2 t_2. \end{aligned}$$

Explicit state form:

$$\begin{aligned} \dot{u} &= vr + \frac{X_1 + X_2 - \frac{1}{2} \rho S C_x u^2}{m}, \\ \dot{v} &= -ur + \frac{Y_1 + Y_2}{m}, \\ \dot{r} &= \frac{Y_1 a_1 - Y_2 a_2 + \Delta X_1 t_1 + \Delta X_2 t_2}{J_z}. \end{aligned}$$

4.2 Planar 7DOF wheel-spin equations

For each wheel:

$$J_{w,ij} \dot{\omega}_{ij} = T_{ij} - R_i F_{xij}.$$

Expanded:

$$\begin{aligned} J_{w,11} \dot{\omega}_{11} &= T_{11} - R_1 F_{x11}, \\ J_{w,12} \dot{\omega}_{12} &= T_{12} - R_1 F_{x12}, \\ J_{w,21} \dot{\omega}_{21} &= T_{21} - R_2 F_{x21}, \\ J_{w,22} \dot{\omega}_{22} &= T_{22} - R_2 F_{x22}. \end{aligned}$$

If drive and brake torques are separated:

$$T_{ij} = T_{d,ij} - T_{b,ij}.$$

So:

$$J_{w,ij} \dot{\omega}_{ij} = T_{d,ij} - T_{b,ij} - R_i F_{xij}.$$

4.3 Planar 7DOF force aggregation

Same as Chapter 9:

$$\begin{aligned} X_{ij} &= F_{xij} \cos \delta_{ij} - F_{yij} \sin \delta_{ij}, \\ Y_{ij} &= F_{xij} \sin \delta_{ij} + F_{yij} \cos \delta_{ij}, \\ X_1 &= X_{11} + X_{12}, & X_2 &= X_{21} + X_{22}, \\ Y_1 &= Y_{11} + Y_{12}, & Y_2 &= Y_{21} + Y_{22}, \\ \Delta X_1 &= \frac{X_{12} - X_{11}}{2}, \\ \Delta X_2 &= \frac{X_{22} - X_{21}}{2}. \end{aligned}$$

4.4 Planar 7DOF longitudinal load transfer

From the Chapter 9 load-transfer equation with $\ddot{\theta} = 0$:

$$\Delta Z = -\frac{Xh}{l}.$$

Since

$$X = ma_x, \quad a_x = \dot{u} - vr,$$

we get

$$\Delta Z = -\frac{ma_x h}{l}.$$

Therefore:

$$\begin{aligned}Z_1 &= \frac{mga_2}{l} - \frac{ma_xh}{l} - Z_1^a, \\Z_2 &= \frac{mga_1}{l} + \frac{ma_xh}{l} - Z_2^a.\end{aligned}$$

4.5 Planar 7DOF lateral load transfer

The most faithful algebraic planar reduction from Chapter 9 keeps the possible $J_{zx}\dot{r}$ contribution:

$$\begin{aligned}\Delta Z_1t_1 &= Y_1q_1 + \lambda_1\left[Y(h-d) + J_{zx}\dot{r}\right], \\ \Delta Z_2t_2 &= Y_2q_2 + \lambda_2\left[Y(h-d) + J_{zx}\dot{r}\right].\end{aligned}$$

Equivalently:

$$\begin{aligned}\Delta Z_1 &= \frac{Y_1q_1 + \lambda_1\left[Y(h-d) + J_{zx}\dot{r}\right]}{t_1}, \\ \Delta Z_2 &= \frac{Y_2q_2 + \lambda_2\left[Y(h-d) + J_{zx}\dot{r}\right]}{t_2}.\end{aligned}$$

Since

$$Y = Y_1 + Y_2 = m(\dot{v} + ur),$$

you may also write

$$Y = ma_y.$$

If the $J_{zx}\dot{r}$ contribution is neglected:

$$\begin{aligned}\Delta Z_1 &\simeq \frac{Y_1q_1 + \lambda_1Y(h-d)}{t_1}, \\ \Delta Z_2 &\simeq \frac{Y_2q_2 + \lambda_2Y(h-d)}{t_2}.\end{aligned}$$

The total lateral load-transfer check is:

$$\Delta Z_1t_1 + \Delta Z_2t_2 = Yh + J_{zx}\dot{r}.$$

If $J_{zx}\dot{r}$ is neglected:

$$\Delta Z_1t_1 + \Delta Z_2t_2 \simeq Yh.$$

4.6 Planar 7DOF individual tyre vertical loads

Using the more complete lateral load transfer with $J_{zx}\dot{r}$:

$$\begin{aligned}Z_{11} &= \frac{1}{2}\left[\frac{mga_2 - ma_xh}{l} - Z_1^a\right] - \frac{Y_1q_1 + \lambda_1\left[Y(h-d) + J_{zx}\dot{r}\right]}{t_1}, \\ Z_{12} &= \frac{1}{2}\left[\frac{mga_2 - ma_xh}{l} - Z_1^a\right] + \frac{Y_1q_1 + \lambda_1\left[Y(h-d) + J_{zx}\dot{r}\right]}{t_1}, \\ Z_{21} &= \frac{1}{2}\left[\frac{mga_1 + ma_xh}{l} - Z_2^a\right] - \frac{Y_2q_2 + \lambda_2\left[Y(h-d) + J_{zx}\dot{r}\right]}{t_2}, \\ Z_{22} &= \frac{1}{2}\left[\frac{mga_1 + ma_xh}{l} - Z_2^a\right] + \frac{Y_2q_2 + \lambda_2\left[Y(h-d) + J_{zx}\dot{r}\right]}{t_2}.\end{aligned}$$

Simplified version with $J_{zx}\dot{r}$ neglected:

$$\begin{aligned}Z_{11} &\simeq \frac{1}{2}\left[\frac{mga_2 - ma_xh}{l} - Z_1^a\right] - \frac{Y_1q_1 + \lambda_1Y(h-d)}{t_1}, \\ Z_{12} &\simeq \frac{1}{2}\left[\frac{mga_2 - ma_xh}{l} - Z_1^a\right] + \frac{Y_1q_1 + \lambda_1Y(h-d)}{t_1}, \\ Z_{21} &\simeq \frac{1}{2}\left[\frac{mga_1 + ma_xh}{l} - Z_2^a\right] - \frac{Y_2q_2 + \lambda_2Y(h-d)}{t_2}, \\ Z_{22} &\simeq \frac{1}{2}\left[\frac{mga_1 + ma_xh}{l} - Z_2^a\right] + \frac{Y_2q_2 + \lambda_2Y(h-d)}{t_2}.\end{aligned}$$

Sanity check:

$$Z_{11} + Z_{12} + Z_{21} + Z_{22} = mg - Z_1^a - Z_2^a.$$

5. Tyre-force closure for both models

For both the corrected Chapter 9 model and the planar 7DOF model, the tyre model closes the equations by giving

$$F_{xij}, F_{yij} = f_{\text{tyre}}\left(\sigma_{xij}, \sigma_{yij}, \gamma_{ij}, Z_{ij}, \dots\right).$$

The vertical load used inside the tyre model is

$$F_{zij} = Z_{ij}.$$

The corrected friction feasibility condition from the errata is

$$X_{ij}^2 + Y_{ij}^2 \leq \mu_p Z_{ij}, \qquad Z_{ij} \geq 0.$$

The errata explicitly corrects this condition in Chapter 3 Eq. (3.122). SV060404a

6. Final compact reference

Corrected Chapter 9 model

$$\begin{aligned}a_x &= (\dot{u} - vr) + (h-d)\ddot{\theta}, \\ a_y &= (\dot{v} + ur) - (h-d)\ddot{\phi}, \\ ma_x &= X_1 + X_2 - \frac{1}{2}\rho SC_x u^2, \\ ma_y &= Y_1 + Y_2, \\ Z_1 + Z_1^a + Z_2 + Z_2^a - mg &= 0, \\ (J_x + m(h-d)^2)\ddot{\phi} - J_{zx}\dot{r} - m(h-d)(\dot{v} + ur) &= -k_\phi\phi - c_\phi\dot{\phi} + mg(h-d)\phi, \\ (J_y + m(h-d)^2)\ddot{\theta} + m(h-d)(\dot{u} - vr) &= -k_\theta\theta - c_\theta\dot{\theta}, \\ J_z\ddot{r} - J_{zx}\ddot{\phi} + m(h-d)\dot{u}\phi &= Y_1a_1 - Y_2a_2 + \Delta X_1t_1 + \Delta X_2t_2, \\ \Delta Z &= -\frac{ma_xh + J_y\ddot{\theta}}{l}, \\ \Delta Z_i &= \frac{Y_iq_i + k_{\phi i}\phi + c_{\phi i}\dot{\phi}}{t_i}, \\ Z_{i1} &= \frac{Z_i}{2} - \Delta Z_i, \qquad Z_{i2} = \frac{Z_i}{2} + \Delta Z_i,\end{aligned}$$

Planar 7DOF model

$$\begin{aligned}a_x &= \dot{u} - vr, \\ a_y &= \dot{v} + ur, \\ \dot{u} &= vr + \frac{X_1 + X_2 - \frac{1}{2}\rho SC_x u^2}{m}, \\ \dot{v} &= -ur + \frac{Y_1 + Y_2}{m}, \\ \dot{r} &= \frac{Y_1a_1 - Y_2a_2 + \Delta X_1t_1 + \Delta X_2t_2}{J_z}, \\ J_{w,ij}\dot{w}_{ij} &= T_{ij} - R_iF_{xij}, \\ \Delta Z &= -\frac{ma_xh}{l}, \\ \Delta Z_i &= \frac{Y_iq_i + \lambda_i\left[Y(h-d) + J_{zx}\dot{r}\right]}{t_i}, \\ Z_{11} &= \frac{1}{2}\left[\frac{mga_2 - ma_xh}{l} - Z_1^a\right] - \Delta Z_1, \\ Z_{12} &= \frac{1}{2}\left[\frac{mga_2 - ma_xh}{l} - Z_1^a\right] + \Delta Z_1, \\ Z_{21} &= \frac{1}{2}\left[\frac{mga_1 + ma_xh}{l} - Z_2^a\right] - \Delta Z_2, \\ Z_{22} &= \frac{1}{2}\left[\frac{mga_1 + ma_xh}{l} - Z_2^a\right] + \Delta Z_2.\end{aligned}$$

For most simplified planar implementations, you may use

$$\Delta Z_i \simeq \frac{Y_iq_i + \lambda_iY(h-d)}{t_i}$$

when $J_{zx}\dot{r}$ is intentionally neglected.

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